

Advances in Global Change Research

Volume 67

Series Editor

Markus Stoffel, Institute of Geological Sciences, University of Geneva, Geneva, Switzerland

Advisory Editors

Wolfgang Cramer, IMEP, Bâtiment Villemin, Europole de l'Arbois,
Aix-en-Provence, France

Urs Luterbacher, University of Geneva, Geneva, Switzerland

F. Toth, International Institute for Applied Systems Analysis (IIASA), Laxanburg,
Austria

More information about this series at <http://www.springer.com/series/5588>

Vincenzo Levizzani • Christopher Kidd
Dalia B. Kirschbaum • Christian D. Kummerow
Kenji Nakamura • F. Joseph Turk
Editors

Satellite Precipitation Measurement

Volume 1

 Springer

Editors

Vincenzo Levizzani
CNR-ISAC
Bologna, Italy

Dalia B. Kirschbaum
Code 617
NASA Goddard Space Flight Center
Greenbelt, MD, USA

Kenji Nakamura
Department of Economics on
Sustainability
Dokkyo University
Saitama, Japan

Christopher Kidd
Earth System Science Interdisciplinary Center
University of Maryland and NASA Goddard
Space Flight Center
Greenbelt, MD, USA

Christian D. Kummerow
Department of Atmospheric Science
Colorado State University
Fort Collins, CO, USA

F. Joseph Turk
Jet Propulsion Laboratory
California Institute of Technology
Pasadena, CA, USA

ISSN 1574-0919

ISSN 2215-1621 (electronic)

Advances in Global Change Research

ISBN 978-3-030-24567-2

ISBN 978-3-030-24568-9 (eBook)

<https://doi.org/10.1007/978-3-030-24568-9>

© Springer Nature Switzerland AG 2020

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Cover illustration: Courtesy of NASA

This Springer imprint is published by the registered company Springer Nature Switzerland AG.
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

*Sognatore è un uomo con i piedi fortemente
appoggiati sulle nuvole**

Ennio Flaiano (1910–1972)

**Dreamer is a man with his feet firmly resting
on the clouds*

In memory of
Arthur Y. Hou (1947–2013)

Preface

This book is published 13 years after the book *Measuring Precipitation from Space: EURAINSAT and the Future* (V. Levizzani, P. Bauer, and F. J. Turk, Eds., Springer, ISBN 978-1-4020-5835-6), but it is not a revised edition of the previous. It is a new book that aims to construct a quasi-complete picture of the science and applications of satellite-derived precipitation measurements at the present time.

The book comes out at the end of a very exciting era of precipitation measurements from space. The Tropical Rainfall Measuring Mission (TRMM), launched in November 1997, ended its long life in space in April 2015 providing an unprecedented 17-year-long dataset of tropical precipitation and lightning. The Global Precipitation Measurement (GPM) mission, launched in February 2014, is now in space as TRMM's natural successor with a more global perspective that extends precipitation radar observations to the Arctic and Antarctic circles. At the same time, the CloudSat mission, launched in April 2006, is in its 13th year in space and focuses on cloud structure, which is essential for improving precipitation retrievals. These are just a few examples of precipitation-oriented missions that continuously provide data from geostationary and low Earth orbits in a truly cooperative effort worldwide. This effort involves many agencies and a broad range of countries who collaborate in a genuine way to observe global precipitation.

It is by realizing the significance of this historical moment and the need to think about what is important for the future that the community joined in the effort of writing a book with the goal of serving the precipitation community itself, the scholars, the students, the stakeholders, the end users, and all the readers interested in knowing the progress of satellite precipitation studies. The most recent achievements in precipitation monitoring from space drive us into the future of measuring not only heavy rainfall but less intense rainfall, snowfall, and even hailfall. Such a scientific framework would not have even been conceivable 13 years ago and is only possible thanks to the relentless effort of the worldwide space and precipitation communities.

Naturally, we realize that at the time of the printing of this book, the field will already have made advances and thus part of the material may already be a bit

outdated. However, in this era of rapidly evolving technological developments, sensors that take years to design, build, and launch are already considered old. This is particularly true nowadays when the progress in approaching new scientific challenges is particularly fast.

Since 2007, science has made substantial progresses toward transforming satellite rainfall “estimates” into accurate “measurements” and producing operational rainfall products readily available for a wide field of applications ranging from climate research and numerical weather prediction to hydrology, agriculture, health, civil protection, and much more. Satellite-derived precipitation products are now being considered as a valuable tool for a number of applications that benefit society and save lives. This is perhaps the most important achievement of all.

This book represents a significant effort, and each author has provided high-quality material in the topics of current and future mission contributions, observations of precipitation using the suite of precipitation satellites, retrieval techniques, validation, and applications. The result is a book that not only photographs the state of the art of the discipline but also projects it into the future.

Bologna, Italy
Greenbelt, MD, USA
Greenbelt, MD, USA
Fort Collins, CO, USA
Saitama, Japan
Pasadena, CA, USA
9 March 2020

Vincenzo Levizzani
Christopher Kidd
Dalia B. Kirschbaum
Christian D. Kummerow
Kenji Nakamura
F. Joseph Turk

Acknowledgments

The first acknowledgment goes to Springer Nature for asking us to start this project and for being very patient with us for the considerable amount of time it took to put the material together.

All the colleagues who spent their precious time contributing their ideas and results deserve special gratitude. They are all very busy scientists, and this is why their contribution is particularly valuable. We deem the book to be a first-hand image of the achievements of the whole community at this time while also providing an important glimpse into future developments.

Then we feel that we need to thank the readers who have already made the previous 2007 Springer book a success, thus de facto making it possible to start writing the new one. We hope you will get from this new book even more inspiration than you got from its predecessor. While some concepts and details will surely become outdated as time goes by, it is our hope that the material contained herein is sufficiently broad that it will always serve as a springboard to understand and put into context the latest research and findings.

It would be almost impossible to thank all the people and organizations behind this effort. You realize this simple truth by looking at the list of contributors and seeing the very long list of institutes, research organizations, university departments, and operational agencies that allowed their members to spend a substantial amount of time writing and correcting the chapters of the book. We thank, in particular, our home institutions that were very supportive in understanding the importance of our work for the community: CNR, Colorado State University, Dokkyo University, JPL-Caltech, NASA, and University of Maryland.

It is very important to remember all the colleagues who are no longer with us and who worked very hard until the last minute providing an essential contribution. This book is dedicated to the memory of a friend of all of us, Arthur Y. Hou (1947–2013). Arthur was not only the US Project Scientist of the Global Precipitation Measurement (GPM) mission, he was a man of a truly global vision who now is in place with the GPM constellation. More than that, he made great efforts to establish an international science cooperation through his gentle and unique way of approaching

each one of us. Other colleagues left us in recent times, and we want to honor them as well: David (Dave) H. Staelin (1938–2011), David I. F. Grimes (1951–2011), and James (Jim) A. Weinman (1930–2012). They all left us much too soon, and we miss them, but their work is here to testify to their essential contribution to the advancement of science and to meet the needs of mankind.

Two major international organizations gave us the opportunity to work together with a global strategy for the future: the International Precipitation Working Group (IPWG) and the World Meteorological Organization (WMO).

The senior editor (Vincenzo Levizzani) would like to recognize the ceaseless work of his coeditors in effectively putting together the material of their respective sections: F. Joseph (Joe) Turk for Section 1, Christian (Chris) D. Kummerow for Sections 2 and 3, Christopher (Chris) Kidd for Section 4, Kenji Nakamura for Section 5, and Dalia B. Kirschbaum for Section 6. Their commitment and competence largely influenced the quality level of this book.

Finally, our families are part of the project through their understanding and their moral and practical support. Without them, the writing of this book would have never even started.

Bologna, Italy
Greenbelt, MD, USA
Greenbelt, MD, USA
Fort Collins, CO, USA
Saitama, Japan
Pasadena, CA, USA
9 March 2020

Vincenzo Levizzani
Christopher Kidd
Dalia B. Kirschbaum
Christian D. Kummerow
Kenji Nakamura
F. Joseph Turk

Contents of Volume 1

Part I Status of Observations and Satellite Programs

1	The Global Precipitation Measurement (GPM) Mission	3
	Christopher Kidd, Yukari N. Takayabu, Gail M. Skofronick-Jackson, George J. Huffman, Scott A. Braun, Takuji Kubota, and F. Joseph Turk	
2	Status of the CloudSat Mission	25
	Matthew D. Lebsock, Tristan S. L’Ecuyer, Norman B. Wood, John M. Haynes, and Mark A. Smalley	
3	The Megha-Tropiques Mission After Seven Years in Space	45
	Rémy Roca, Michel Dejus, Philippe Chambon, Sophie Cloché, and Michel Capderou	
4	Microwave Sensors, Imagers and Sounders	63
	Kazumasa Aonashi and Ralph R. Ferraro	
5	Microwave and Sub-mm Wave Sensors: A European Perspective	83
	Christophe Accadia, Vinia Mattioli, Paolo Colucci, Peter Schlüssel, Salvatore D’Addio, Ulf Klein, Tobias Wehr, and Craig Donlon	
6	Plans for Future Missions	99
	Christian D. Kummerow, Simone Tanelli, Nobuhiro Takahashi, Kinji Furukawa, Marian Klein, and Vincenzo Levizzani	

Part II Retrieval Techniques, Algorithms and Sensors

7	Introduction to Passive Microwave Retrieval Methods	123
	Christian D. Kummerow	

8	The Goddard Profiling (GPROF) Precipitation Retrieval Algorithm	141
	David L. Randel, Christian D. Kummerow, and Sarah Ringerud	
9	Precipitation Estimation from the Microwave Integrated Retrieval System (MiRS)	153
	Christopher Grassotti, Shuyan Liu, Quanhua Liu, Sid-Ahmed Boukabara, Kevin Garrett, Flavio Iturbide-Sanchez, and Ryan Honeyager	
10	Introduction to Radar Rain Retrieval Methods	169
	Toshio Iguchi and Ziad S. Haddad	
11	Dual-Frequency Precipitation Radar (DPR) on the Global Precipitation Measurement (GPM) Mission's Core Observatory . . .	183
	Toshio Iguchi	
12	DPR Dual-Frequency Precipitation Classification	193
	V. Chandrasekar and Minda Le	
13	Triple-Frequency Radar Retrievals	211
	Alessandro Battaglia, Simone Tanelli, Frederic Tridon, Stefan Kneifel, Jussi Leinonen, and Pavlos Kollias	
14	Precipitation Retrievals from Satellite Combined Radar and Radiometer Observations	231
	Mircea Grecu and William S. Olson	
15	Scattering of Hydrometeors	249
	Stefan Kneifel, Jussi Leinonen, Jani Tyynelä, Davide Ori, and Alessandro Battaglia	
16	Radar Snowfall Measurement	277
	Guosheng Liu	
17	A 1DVAR-Based Snowfall Rate Algorithm for Passive Microwave Radiometers	297
	Huan Meng, Cezar Kongoli, and Ralph R. Ferraro	
18	X-Band Synthetic Aperture Radar Methods	315
	Saverio Mori, Frank S. Marzano, and Nazzareno Pierdicca	

Part III Merged Precipitation Products

19	Integrated Multi-satellite Retrievals for the Global Precipitation Measurement (GPM) Mission (IMERG)	343
	George J. Huffman, David T. Bolvin, Dan Braithwaite, Kuo-Lin Hsu, Robert J. Joyce, Christopher Kidd, Eric J. Nelkin, Soroosh Sorooshian, Erich F. Stocker, Jackson Tan, David B. Wolff, and Pingping Xie	

20 Global Satellite Mapping of Precipitation (GSMaP) Products in the GPM Era 355
Takuji Kubota, Kazumasa Aonashi, Tomoo Ushio, Shoichi Shige, Yukari N. Takayabu, Misako Kachi, Yoriko Arai, Tomoko Tashima, Takeshi Masaki, Nozomi Kawamoto, Tomoaki Mega, Munehisa K. Yamamoto, Atsushi Hamada, Moeka Yamaji, Guosheng Liu, and Riko Oki

21 Improving PERSIANN-CCS Using Passive Microwave Rainfall Estimation 375
Kuo-Lin Hsu, Negar Karbalee, and Dan Braithwaite

22 TAMSAT 393
Ross Maidment, Emily Black, Helen Greatrex, and Matthew Young

23 Algorithm and Data Improvements for Version 2.1 of the Climate Hazards Center’s InfraRed Precipitation with Stations Data Set 409
Chris Funk, Pete Peterson, Martin Landsfeld, Frank Davenport, Andreas Becker, Udo Schneider, Diego Pedreros, Amy McNally, Kristi Arsenault, Laura Harrison, and Shraddhanand Shukla

24 Merging the Infrared Fleet and the Microwave Constellation for Tropical Hydrometeorology (TAPEER) and Global Climate Monitoring (GIRAFE) Applications 429
Rémy Roca, Adrien Guérou, Rômulo A. Jucá Oliveira, Philippe Chambon, Marielle Gosset, Sophie Cloché, and Marc Schröder

Contents of Volume 2

Part IV Validation

25	The IPWG Satellite Precipitation Validation Effort	453
	Christopher Kidd, Shoichi Shige, Daniel Vila, Elena Tarnavsky, Munehisa K. Yamamoto, Viviana Maggioni, and Bathobile Maseko	
26	The GPM Ground Validation Program	471
	Walter A. Petersen, Pierre-Emmanuel Kirstetter, Jianxin Wang, David B. Wolff, and Ali Tokay	
27	The GPM DPR Validation Program	503
	Riko Oki, Toshio Iguchi, and Kenji Nakamura	
28	Error and Uncertainty Characterization	515
	Christian Massari and Viviana Maggioni	
29	Multiscale Evaluation of Satellite Precipitation Products: Effective Resolution of IMERG	533
	Clément Guilloteau and Efi Foufoula-Georgiou	
30	Remote Sensing of Orographic Precipitation	559
	Ana P. Barros and Malarvizhi Arulraj	
31	Integrated Multi-satellite Evaluation for the Global Precipitation Measurement: Impact of Precipitation Types on Spaceborne Precipitation Estimation	583
	Pierre-Emmanuel Kirstetter, Walter A. Petersen, Christian D. Kummerow, and David B. Wolff	
32	Hydrologic Validation and Flood Analysis	609
	Witold F. Krajewski, Felipe Quintero, Mohamed El Saadani, and Radoslaw Goska	

33	Global-Scale Evaluation of 22 Precipitation Datasets Using Gauge Observations and Hydrological Modeling	625
	Hylke E. Beck, Noemi Vergopolan, Ming Pan, Vincenzo Levizzani, Albert I. J. M. van Dijk, Graham P. Weedon, Luca Brocca, Florian Pappenberger, George J. Huffman, and Eric F. Wood	
34	OceanRAIN – The Global Ocean Surface-Reference Dataset for Characterization, Validation and Evaluation of the Water Cycle	655
	Christian Klepp, Paul A. Kucera, Jörg Burdanowitz, and Alain Protat	

Part V Observed Characteristics of Precipitation

35	GPCP and the Global Characteristics of Precipitation	677
	Robert F. Adler, Guojun Gu, George J. Huffman, Mathew R. P. Sapiano, and Jian-Jian Wang	
36	Global Snowfall Detection and Measurement	699
	Mark S. Kulie, Lisa Milani, Norman B. Wood, and Tristan S. L'Ecuyer	
37	Snowfall Detection by Spaceborne Radars	717
	Atsushi Hamada, Toshio Iguchi, and Yukari N. Takayabu	
38	On the Duration and Life Cycle of Precipitation Systems in the Tropics	729
	Rémy Roca, Dominique Bouniol, and Thomas Fiolleau	
39	Observational Characteristics of Warm-Type Heavy Rainfall	745
	Byung-Ju Sohn, Geun-Hyeok Ryu, and Hwan-Jin Song	
40	Satellite Precipitation Measurement and Extreme Rainfall	761
	Olivier P. Prat and Brian R. Nelson	
41	Rainfall Trends in East Africa from an Ensemble of IR-Based Satellite Products	791
	Elsa Cattani, Andrés Merino, and Vincenzo Levizzani	
42	Heavy Precipitation Systems in the Mediterranean Area: The Role of GPM	819
	Giulia Panegrossi, Anna Cinzia Marra, Paolo Sanò, Luca Baldini, Daniele Casella, and Federico Porcù	
43	Dryland Precipitation Climatology from Satellite Observations	843
	Efrat Morin, Francesco Marra, and Moshe Armon	
44	Hailfall Detection	861
	Ralph R. Ferraro, Daniel Cecil, and Sante Laviola	
45	Improving High-Latitude and Cold Region Precipitation Analysis	881
	Ali Behrangi	

- 46 Latent Heating Retrievals from Satellite Observations 897**
Yukari N. Takayabu and Wei-Kuo Tao

Part VI Applications

- 47 Operational Applications of Global Precipitation Measurement Observations 919**
Anita LeRoy, Emily Berndt, Andrew Molthan, Bradley Zavodsky, Matthew Smith, Frank LaFontaine, Kevin McGrath, and Kevin Fuell
- 48 Assimilation of Precipitation Observations from Space into Numerical Weather Prediction (NWP) 941**
Sid-Ahmed Boukabara, Erin Jones, Alan Geer, Masahiro Kazumori, Kevin Garrett, and Eric Maddy
- 49 Precipitation Ensemble Data Assimilation in NWP Models 983**
Takemasa Miyoshi, Shunji Kotsuki, Koji Terasaki, Shigenori Otsuka, Guo-Yuan Lien, Hisashi Yashiro, Hirofumi Tomita, Masaki Satoh, and Eugenia Kalnay
- 50 PERSIANN-CDR for Hydrology and Hydro-climatic Applications 993**
Phu Nguyen, Hamed Ashouri, Mohammed Ombadi, Negin Hayatbini, Kuo-Lin Hsu, and Soroosh Sorooshian
- 51 Soil Moisture and Precipitation: The SM2RAIN Algorithm for Rainfall Retrieval from Satellite Soil Moisture 1013**
Luca Ciabatta, Stefania Camici, Christian Massari, Paolo Filippucci, Sebastian Hahn, Wolfgang Wagner, and Luca Brocca
- 52 Drought Risk Management Using Satellite-Based Rainfall Estimates 1029**
Elena Tarnavsky and Rogerio Bonifacio
- 53 Two Decades of Urban Hydroclimatological Studies Have Yielded Discovery and Societal Benefits 1055**
J. Marshall Shepherd, Steven J. Burian, Menglin Jin, Chuntao Liu, and Bradford Johnson
- 54 Validation of Climate Models 1073**
Francisco J. Tapiador
- 55 Extreme Precipitation in the Himalayan Landslide Hotspot 1087**
Thomas Stanley, Dalia B. Kirschbaum, Salvatore Pascale, and Sarah Kapnick
- 56 The Value of Satellite Rainfall Estimates in Agriculture and Food Security 1113**
Tufa Dinku

57	Using Satellite Estimates of Precipitation for Fire Danger Rating	1131
	Robert D. Field	
58	Variability of Satellite Sea Surface Salinity Under Rainfall	1155
	Alexandre Supply, Jacqueline Boutin, Gilles Reverdin, Jean-Luc Vergely, and Hugo Bellenger	

List of Figures

Fig. 1.1	Schematic of the Global Precipitation Measurement (GPM) mission Core Observatory (CO; left) and the GPM international partner constellation (right). Note that as of 21 May 2018 the KaPR swath width has been increased to 245 km to match that of the KuPR. (Note that the GPM-CO alternates flight directions to keep the canted solar panel towards the Sun: half the time the flight direction is 180° from that shown)	5
Fig. 1.2	An infrared image from Himawari 8 of the “lake effect” clouds over the Japan Sea (top), and three-dimensional snapshot of shallow precipitation from those clouds observed by the effective radar reflectivity at Ku band of the GPM/DPR at 0955 UTC 2 December 2014 (bottom)	7
Fig. 1.3	Hail detection in the thunderstorm near Fort Worth, Texas, on 26 May 2015. (a) Hydrometeor identification by a ground-based polarimetric radar and (b) the output from the “flagHeavyIcePrecip” from the DPR product. (Adapted from Iguchi et al. 2018; see the reference for details)	8
Fig. 2.1	Highlights several of the unique features of the CloudSat data. The image shows an example of a tropical deep convective system with obvious heavy attenuation and multiple scattering effects. Attenuation can be so heavy at times that the surface reflection is not observed. The convective core area is identifiable through the lack of a radar bright band and elevated reflectivity maximum. Also notice the frequent detection of shallow isolated light showers with reflectivity generally <8 dBZ	27

Fig. 2.2	Panel (a) shows an estimate of effect of CloudSat fixed diurnal sampling on the annual mean Probability of Precipitation (PoP). The map shows the difference of the full CMORPH dataset from 2007 to 2010 from the CMORPH dataset subsampled at the CloudSat ground track. In most regions the effect causes an underestimate of the PoP that can be as large as 6%. Panel (b) shows an estimate of the effect of non-scanning sampling on the PoP again using CMORPH. The CMORPH data is restricted to the CloudSat sampling times at each latitude and compared to that calculated from a single cross section at a random longitude	28
Fig. 2.3	The fraction of total (rain & no-rain) pixels in which the surface signal is saturated by heavy attenuation	28
Fig. 2.4	The frequency of occurrence of surface precipitation of any phase. Occurrence is estimated using the certain precipitation flag in the 2C-Precip-Column product described below using data from 2007 to 2010	29
Fig. 2.5	The global frequency of occurrence of convective cores identified in CloudSat’s 2C-Precip-Column product using data from 2007 to 2010	31
Fig. 2.6	Variation of clear-sky surface backscatter with wind speed as derived from matched AMSR-E and CloudSat wind observations over the ocean, for a multi-month period and a fixed sea surface temperature range of 15 to 25 °C. Colors indicate normalized frequency of occurrence; the red line is the mean, and the dashed black lines are one standard deviation either side of the mean ...	34
Fig. 2.7	(panel a) the accumulate rainfall from the CloudSat 2C-Rain Profile product for the years 2007–2010; (panel b) the same from the DRP Ku band algorithm for a 2-year period from 2014 to 2016; (panel c) the difference between panels a and b; (panel d) the scatter plot of the rate difference shown in panel c with the saturation occurrence fraction from CloudSat	40
Fig. 3.1	Two-years times series of the flip maneuver schedule based on the beta solar angle (the angle between the sun direction and the orbital plan; see Capderou 2014 for details)	48
Fig. 3.2	Schematic of the effect of the relaxed control of the yaw	48
Fig. 3.3	Time series of maximum (red) and minimum (blue) latitude of the scan of the SAPHIR instrument	50
Fig. 3.4	Time series of the noise (NEΔT) for each channel of the SAPHIR sounder	51
Fig. 3.5	Zonal mean of the fraction of time for which the baseline product and the No-meghatropiques products differs by more than 50% of the daily accumulation. Summer 2012 conditions are considered. (Adapted from Roca et al. 2018)	52

Fig. 3.6	Brightness temperatures from SAPHIR channel 6 over the Tropical Atlantic on 6 September, 2017. The four scans correspond to consecutive orbits and show the developments of the two hurricanes IRMA and JOSE. These two hurricanes were observed within the same orbit, illustrating the unique feature of this observing system	54
Fig. 3.7	Fraction of SAPHIR observations per month which have been received and used a Météo-France before the cutoff times of the Météo-France global data assimilation system (4–5 h depending on the assimilation cycle) (Courtesy of Hervé Benichou, Météo-France DIROP/COMPAS/COM). The period starts in June 2015, which corresponds to the beginning of the operational assimilation of Megha-Tropiques data at Météo-France. The dashed red line refers to the averaged fraction over this 4-year period	56
Fig. 4.1	A time sequence of the microwave images of the super-typhoon NANGKA from 0540 to 1731 UTC 12 July 2015, superimposed on MTSAT IR images. (From NRL Tropical Cyclone page – https://www.nrlmry.navy.mil/TC.html , last accessed 21 Oct 2018)	64
Fig. 4.2	Scan geometries of the SSM/I sensors. (The COMET Program – https://www.meted.ucar.edu/index.php , last accessed 21 Oct. 2018)	68
Fig. 4.3	Scan geometries of the TRMM PR, TMI and VIRS. (Adapted from Kummerow et al. 1998)	69
Fig. 4.4	Scan geometries of the AMSU-A and AMSU-B sensors. (The COMET Program – https://www.meted.ucar.edu/index.php , last accessed 21 Oct 2018)	76
Fig. 5.1	MWI and ICI accommodation on Metop-SG	84
Fig. 6.1	State-of-the-art microwave monolithic integrated circuits low noise amplifiers' noise figure	108
Fig. 6.2	A deployable reflector antenna concept for the small satellites. The antenna reflector surface can fit to less than 1.5 U volume and it is designed to operate up to ~100 GHz	108
Fig. 6.3	Roadmap of spaceborne precipitation radar	112
Fig. 7.1	Demonstrative atmospheric transmittances (total, H ₂ O and O ₂) as a function of frequency and wavelength in the microwave region. (Adapted from Liou 2002, p. 415)	125
Fig. 7.2	Example of brightness temperatures as a function of column averaged rain rate at 19 GHz used to illustrate the “beamfilling” or non-homogeneous rain distribution effect	127

Fig. 8.1	Schematic for the GPROF Processing Algorithm. The three main components are the Sensor Profile Database, the Preprocessor, and the GPROF 2017 Processing Engine	143
Fig. 8.2	GPROF database distribution of Total Precipitable Water (TPW) and Two-Meter Temperature (T2M) profiles included in the a priori database year. These represent the number of database profiles in each TPW/T2 m and Surface Type category	146
Fig. 8.3	GPROF GMI retrieval of surface rain for hurricane Harvey, 25 August 2017 shortly before making landfall on the Texas coast. Over 60 inches (1500 mm) of precipitation was recorded near Houston over the next 5 days	149
Fig. 8.4	Solid (Snow) vs. Liquid Precipitation and Wet Bulb for both ocean and land locations. (Adapted from Sims and Liu 2015)	150
Fig. 8.5	GPROF zonal averaged precipitation retrieval for Total (left) and Frozen precipitation (right) for five GPM constellation sensors	151
Fig. 9.1	Schematic of MiRS processing components and data flow showing MiRS core retrieval and post-processing components. Core products are retrieved simultaneously as part of the state vector. Post-processing products are derived through vertical integration (water vapor, hydrometeors), catalogs (SIC, SWE), or fast regressions (rain rate). Post-processed hydrometeor retrieval products are indicated in red: Rain Rate, Graupel Water Path, Rain Water Path and Cloud Liquid Water	155
Fig. 9.2	Example of rain water (left) and graupel water (right) retrieval evolution for a single vertical profile based on NOAA-18 AMSU-MHS measurements. Top panels show rain and graupel water profile retrieval as function of iteration (3 iterations total). The remaining panels show the CRTM Jacobians with respect to rain and graupel at channels 15, 17, 18, 19, 20 (89, 157, 183 ± 1 , 183 ± 3 , and 190 GHz), for each iteration. In this case, the retrieval converged in 3 iterations. Rain and graupel particle effective radii were assumed to be 500 microns	159
Fig. 9.3	Comparison of global rain rate maps on 21 June 2016 from MiRS when applied to GPM/GMI (left) and SNPP/ATMS measurements (right). Examples of weather systems detected by both satellites are circled	160

Fig. 9.4	MiRS retrievals of hydrometeor and temperature structure around Typhoon Soudelor from Suomi-NPP/ATMS valid 0445 UTC on 6 August 2015. Panels show surface rain rate (top left), rain water 0.01 mm isosurface with temperature profile superimposed (top right), graupel water 0.05 mm isosurface with temperature profile superimposed (bottom left), and a vertical cross-section along 21°N of both rain and graupel water (bottom right)	161
Fig. 9.5	Comparison of MiRS SNPP/ATMS instantaneous rain rate (mm/h) (top) with operational NWS Stage IV rain rate (bottom) over the conterminous US for two different dates, 16 March 2016 (left), and 28 July 2016 (right)	162
Fig. 9.6	Example of impact of using retrieved CLW over land in the land precipitation estimation from SNPP/ATMS on 01 May 2016. Shown are (a) MiRS operational rain rate (mm/h), (b) MiRS rain rate using CLW, (c) MRMS Q3 radar-gauge analysis valid at 1900 UTC (units in inches), (d) MiRS Liquid Water Path (LWP = RWP + CLW, mm), and (e) visible satellite image from GOES-East valid at 1915 UTC	164
Fig. 9.7	Probability distribution functions of MIRS ATMS vs. Stage IV baseline (operational, no CLW included in rain rate estimation) and experimental rain rate (CLW included) over land during September–November 2016. Note improved frequency distribution and agreement with Stage IV in experimental rain rate. Distributions are for all points with Stage IV rain rate greater than 0 mm h ⁻¹	165
Fig. 11.1	DPR's scan pattern before May 21 2018 (left) and after May 21 2018 (right). KaHS beams scan in the inner swath before May 21 2018, but now they scan in the outer swath and match with KuPR's beams. Numbers in color indicate angle bin numbers for KuPR (blue), KaMS (yellow), and KaHS (red)	184
Fig. 11.2	DPR L2 algorithm flow	186
Fig. 12.1	Schematic plot of DFR _m profile with key points A, B, C, and D. Point A: slope of DFR _m has peak value. Point B: local maximum of DFR _m . Point C: local minimum of DFR _m . Point D: DFR _m value near surface	194
Fig. 12.2	Histogram of DFR _m index V3 and CDFs (cumulative density function) using total of 121,859 vertical profiles from GPM real data. (a) Histogram and 1-CDF of V3 for Stratiform rain. Red dashed line represents 1-CDF. (b) Histogram and CDF of V3 for Convective rain. Red dashed line represents CDF	196
Fig. 12.3	Block diagram of precipitation type classification model	197
Fig. 12.4	Block diagram of melting layer detection for DFR _m method	198

Fig. 12.5	Left column (from top to bottom): comparison of melting layer top height (in km) between dual-frequency classification method and Ku only method for cyclone, hurricane and typhoons shown in Table 12.2. Right column illustrates similar results for melting layer bottom height (in km) 200	200
Fig. 12.6	(a) GPM DPR overpass of rainfall rate on March 17, 2014 (#000272). Circled A, B and C represents snow, stratiform rain, and convective rain. (b) Averaged reflectivity profiles as well as dual-frequency ratio profile for snow. (c) Same as (b) for stratiform rain. (d) Same as (b) for convective rain 202	202
Fig. 12.7	Vertical cross section at nadir of DPR overpass shown in Fig. 12.6a 203	203
Fig. 12.8	GPM DPR overpass of rainfall rate on March 17, 2014. Scan # from 4894 to 5142. (a) Histogram of mean DFR_m slope in absolute value. (b) Histogram of maximum reflectivity at Ku band. (c) Histogram of storm top height 204	204
Fig. 12.9	Large scale study of the snow index using GPM DPR profiles. Histograms of the snow index are shown for rain (blue) and snow (red). The blue dashed curve is the cumulative density function (CDF) for rain. The red dashed curve is 1-CDF for snow 205	205
Fig. 12.10	Flowchart to perform surface snowfall identification in profile classification module of GPM DPR level 2 algorithm 206	206
Fig. 12.11	Match ratio for 16 validation cases during the years 2014–2018 208	208
Fig. 13.1	CloudSat and GPM coincident overpass observations of a convective precipitation system developed over the Banda Sea in the Maluku Islands of Indonesia. Measurements from a suite of microwave sensors are shown. Top row: CloudSat W-band reflectivity; second row: GPM Ka-band reflectivities for the high sensitivity (HS) scan; third row: GPM Ku-band reflectivity for the normal scan (NS). The dataset of coincident overpasses is from the GPM product 2B-CSATGPM from the NASA Precipitation Processing System developed by J. Turk, JPL 213	213
Fig. 13.2	Left: effective DFR_{Ka-W} vs DFR_{Ku-Ka} for population of raindrops at 15 °C with Γ DSDs with different m as indicated in the legend and with color-coded mean mass-weighted diameter. Right: extinction coefficient vs rain rate for exponential DSDs with intercept parameters as indicated in the legend. Scattering properties are computed using T-matrix 216	216
Fig. 13.3	Effective DFR_{Ka-W} vs DFR_{Ku-Ka} for population of ice crystals computed different scattering tables. The density plots show the distribution measured by a triple-frequency radar during one of the GPM field campaigns 218	218

Fig. 13.4	Left: gas-corrected Ka-band reflectivity (top), DFR_{Ku-Ka} (center) and DFR_{Ka-W} (bottom) for a flight on the 1 December 2016. Right top: flight tracks across the Olympic Peninsula from the Olympic Mountains range toward and beyond the NPOL radar (black dot) on the Pacific coastline. The UTC time of DC-8 (external contour) and Citation (internal contour) aircraft paths are modulated in color (see color bar). The position of the Citation is shown in the top left panel. Right bottom: hydrometeor classification according to the multi-frequency method of Tridon et al. (2019)	220
Fig. 13.5	Retrieved parameters for the leg shown in Fig. 13.4: mean mass-weighted maximum size (top), IWC (center) and flux (bottom). The right bottom panel shows the bulk ice density as defined in Leinonen et al. (2018)	221
Fig. 13.6	Comparison between in-situ and retrieved microphysical properties as sample by the Citation aircraft in the ice part corresponding to the upper leg as shown in Fig. 13.4. Left column: mean mass weighted particle size (top) and ice water content (bottom) for the in-situ and for the two retrievals considered in this paper. The blue lines and the blue bands correspond to the a-priori and its standard deviation. Right panel: scatterplot of IWC vs D_m for the in-situ and for the two retrievals. Note that the in-situ D_m and IWC are derived from the PSD measurements based on the assumption that the mass-size follows that of lightly rimed B-model of Leinonen and Szyrmer (2015)	222
Fig. 13.7	Shift of the retrieved particle size distributions towards lower sizes. Figure prepared by L. Pfizenmaier (see Pfizenmaier et al. 2019 for details)	224
Fig. 14.1	Illustration of two possible strategies to mitigate the large mismatches between the radar and radiometer footprint sizes in satellite combined retrievals	234
Fig. 14.2	(Left) Example of combined surface precipitation estimates for GPM orbit 605 on 1 October 2014. (Right) The associated Multi-Radar/Multi-Sensor (MRMS) estimates	238
Fig. 14.3	Density scatterplot of GPM combined V06 vs reference MRMS convective precipitation (mm h^{-1}) at the footprint scale over the period April 2014–October 2014	239
Fig. 14.4	(Left) Example of liquid water path (LWP) derived from GMI observations over oceans for orbit 003351 on 1 October 2014. (Right) The associated combined surface precipitation. The magnitude of the LWP suggests that light precipitation is undetected by the DPR	240

Fig. 14.5	(Left) Example of Ku-band PIA derived by the combined algorithm for orbit 003539 on 13 October 2014. (Right) The associated Ku-band PIA estimated exclusively from GMI observations	242
Fig. 14.6	(Left) Ku-band PIA derived by the combined algorithm for orbit 003351 on 1 October 2014. (Right) The associated Ku-band PIA derived from GMI brightness temperatures using the extended formulation described in the text	244
Fig. 15.1	Extinction (left) and backscattering (right) cross sections for spheres (continuous lines), perfectly oriented spheroids (dashed) and Rayleigh spheres (dotted) for single raindrops at 9.6 (red), 35.5 (green) and 94 (blue) GHz	260
Fig. 15.2	Reflectivity per unit mass for an exponential drop size distribution vs mean mass-weighted equi-volume diameter for spheres (continuous lines), perfectly oriented spheroids (dashed) at 9.6 (red), 35.5 (green) and 94 (blue) GHz	260
Fig. 15.3	Backscattering (Q_{bk} , left) and extinction (Q_{ext} , right) efficiencies defined as $Q = \sigma / \pi r_{eff}^2$ with r_{eff} being the radius of the equal mass ice sphere and σ the corresponding cross section. The size parameter x combines the dependence of the scattering variables on particle mass and wavelength. Spheroid approximations include spheres of solid ice (continuous black line), spheres with an ice-air mixture (black long-dashed) representing the mass-size relation of Brown and Francis (1995), and spheroids with the same mass-size relation but with and aspect ratio of 0.6 (black dotted). Scattering properties for unrimed (gray dots) and rimed (black dots) calculated with DDA in Leinonen and Szyrmer (2015) (results shown are for particle model B and the second most rimed particles). The SSRGA (gray solid line) has been derived for the same ensemble of unrimed aggregates (Leinonen et al. 2018a, b). The vertical lines denote the size parameter for X, Ka and W Band assuming a particle mass of 10 mg which corresponds for example for the unrimed aggregates to a maximum snowflake size of 3 cm	263
Fig. 15.4	Extinction (left) and backscattering (right) cross sections for partially melted snow aggregates at 10% (dark) and 50% (gray) melted fractions. Results are from DDA scattering simulations (Ori et al. 2014) at 9.6 (continuous), 35.5 (dashed) and 94 (dash-dotted) GHz	269
Fig. 16.1	The spread of Z-S relations under different assumptions of particle shapes. (Adapted from Hiley et al. 2011)	280

Fig. 16.2	Z-S relation for three nonspherical snowflakes. A least square fitting curve and relation by Matrosov (2007) are also shown. (Adapted from Liu 2008b)	281
Fig. 16.3	Scatterplot of coincident CloudSat CPR and GPM/DPR Ku-band reflectivities for snowing cases. Data in the green box are those above the minimum detection for both Ku and CPR radars	282
Fig. 16.4	PDFs and cumulative PDFs of snowfall occurrence and snowfall rate derived by coincident CPR and DPR observations using Z-S relations as discussed in the text. Data period for these plots is from March 2014 to December 2015	283
Fig. 16.5	CloudSat CPR radar reflectivity for 2 snowfall cases on 21 August 2007 (top) and 17 July 2006 (bottom) over 50–60°S. The top case is associated with deep snowing clouds cross a frontal system and the bottom case is associated with very shallow snowing cloud cells next to moderate deep snowing clouds	284
Fig. 16.6	Frequency of total precipitation (rain and snow, left), frequency of snowfall (middle) and mean snowfall rate (right) for northern (top) and southern (bottom) hemispheres derived from CloudSat observations from July 2006 to June 2008. The diagrams cover the area from the Poles to 40°N/S. Detailed descriptions of the retrieval method can be found in Liu (2008b)	285
Fig. 16.7	Zonally averaged frequency of occurrence of rainfall (blue), frequency of snowfall (gray, stacked above blue) and snowfall rate (red). Derived using CPR –15 dBZ as precipitation threshold, rain-snow separation scheme of Sims and Liu (2015), and Z-S relation of Liu (2008b)	286
Fig. 16.8	Number and volume frequency distributions of snowfall rate derived from CloudSat observations. Note the frequencies are calculated in a snowfall rate interval on logarithm scale	286
Fig. 16.9	Mean profiles for a given near surface snowfall rate for (a) over ocean and (b) over land environments. CloudSat observations from July 2006 to June 2007 are used. (Adapted from Liu 2008b)	287
Fig. 16.10	Relation between near surface snowfall rate and cloud top height as expressed by snowfall profiles frequency distributions for snowfall events over (a) ocean and (b) land. The frequency values are normalized so that the maximum frequency is 100. (Adapted from Liu 2008b)	288

Fig. 16.11	Mean liquid water path as a function of cloud top temperature and near surface radar reflectivity for different snowing cloud types: (a) Isolated shallow clouds, (b) Isolated deep clouds, (c) extended shallow clouds, and (d) Extended deep clouds. Coincident CloudSat CPR and Aqua AMSR-E observations from June 2006 to June 2010 are used. Note that the LWP color scales for isolated and extended clouds are different. (Adapted from Wang et al. 2013)	289
Fig. 16.12	Mean snowfall rate maps derived from multiple years of CloudSat (2007–2010), GMI (March 2014–Feb 2018), and MHS (2007–2010). GMI algorithm is trained by combined CloudSat and DPR radar data. No observations in blank areas. MHS algorithm is trained by CloudSat CPR	291
Fig. 17.1	CloudSat derived ice water content (IWC) profiles. The values have been normalized	306
Fig. 17.2	Stage IV vs. S-NPP SFR scatter plot from (a) before calibration, and (b) after calibration	307
Fig. 17.3	PDFs of S-NPP SFR and Stage IV (a) before, and (b) after calibration	308
Fig. 17.4	S-NPP ATMS SFR using (a) satellite-only SD algorithm, and (b) hybrid SD algorithm during a major snowfall event on 5 February 2014 in the US Image (c) is the near coincident radar reflectivity which covers both snowfall and rainfall (in the southern part of CONUS). The noted oval areas in (a) and (b) show legitimate snowfall that was missed by the satellite-only algorithm but captured by the hybrid algorithm	309
Fig. 17.5	(a) Scatter plot of Stage IV vs. collocated S-NPP SFR validation data, and (b) PDFs of the same data sets	310
Fig. 17.6	Comparison of (a) S-NPP SFR 3-month average from January - March 2017 and (b) the corresponding Stage IV data	310
Fig. 18.1	(Lower left image) Synoptic view of Hurricane Gustav over south eastern Louisiana on September 2, 2008 12:00 UTC taken from NEXRAD weather radar reflectivity mosaic. The white box shows an outer rain band around 30.5° N $\times$ 89.5° W. (Central image) Geographic representation of the NEXRAD image at 0.86° elevation, acquired by the S-band radar (KMOB, in figure) near Mobile (Alabama). The semi-transparent rectangular box represents the scene of interest, acquired by TSX X-SAR on 2 September 2008 12:00 UTC in HH polarization and ScanSAR mode (100 km swath). (Upper right image) TSX quicklook of the acquisition in arbitrary units at 100-m resolution; flight direction is indicated. (Adapted from Marzano et al. 2010)	319

Fig. 18.2	Correlation diagram between NRCS values X-band σ SAR against co-located and co-registered S-band NEXRAD weather-radar reflectivity Z , for a selected region of interest (ROI) of the scene. The upper axis provides the estimated rain-rate from NEXRAD data using the Marshall-Palmer relation (Bringi and Chandrasekar 2001). The best-fitting curve is also plotted. (Adapted from Marzano et al. 2010)	320
Fig. 18.3	TRMM observations, at 15:30 UTC, for the case study of Hurricane Gustav. (Top panels) TRMM 1B11 brightness temperature (TB) product relative to TMI channel 7 (37 GHz horizontal polarization, left), beam effective field-of-view (EFOV) of $16 \times 9 \text{ km}^2$, and TMI channel 9 (85.5 GHz horizontal polarization, right) with a main-beam EFOV of $7 \times 5 \text{ km}^2$. The cyclonic cell indicated in Fig. 18.2 is well captured. (Bottom panels) The TRMM 1C21 radar reflectivity (dBZ) product, relative to PR normal sample (left) range bin 75, and PR rain oversample (right), range bin 16. Note that the PR swath is 220 km wide (reduced in the oversampled product) and the range resolution is 0.25 km; TMI swath is 760 km wide. (Adapted from Marzano et al. 2011)	321
Fig. 18.4	Schematic SAR NRCS (in dB) as a function of cross-track scanning distance x , showing enhanced values on the left of the cross-over point caused by scattering from the cloud top and attenuation from rain in the lower cloud on the right. The viewing angle with respect to nadir (incidence angle) is θ , while the cloud extension is w . The symbol Δr indicates the width of the slant slice of the atmosphere representing the SAR side-looking resolution volume. The figure also shows the energy fluxes and the e.m. parameters of the model according to Marzano et al. (2012) and Mori et al. (2017a)	323
Fig. 18.5	Example of System for Atmospheric Model (SAM) vertical slice for a Compact Medium Single Cell cloud. Values indicate water content W in g m^{-3} of the simulated distributions of snow, rain, ice and cloud particles	326
Fig. 18.6	SAR simulated response in terms of normalized Radar cross section $\sigma_{\text{SAR}hh}$ (horizontal transmitted and received), co-polar ratio $Z_{\text{SAR}co}$ and complex correlation coefficient $\rho_{\text{SAR}co}$ for the SAM realistic cell of Fig. 18.5. Four SAR frequencies evaluated (5.4, 9, 14 and 35 GHz). Considered background are Spheres, Dihedrals, Cylinders, and a semi-empirical bare soil (SEM)	327
Fig. 18.7	Flowchart of the procedure described in Mori et al. (2016) for detecting flooded and cloud areas in X-SAR images and estimating the relative precipitation rate	332

Fig. 18.8	Voghiera case study. Left image is a geocoded quicklook of the CSK acquisition (at 05:18 UTC). Right figure shows the corresponding Italian National Mosaic Vertical Maximum Intensities (VMI) at 05:30 UTC (~15 min acquisition time); the ellipse approximately encloses the case study area. (Adapted from Mori et al. 2017b)	334
Fig. 18.9	Precipitation maps for the case study of Fig. 18.8. Left map is obtained from X-SAR data with the procedure of Sect. 18.5.1 (filtered and smoothed to WR resolution), right map is obtained by WR VMI data and a Marshall-Palmer formula. (Adapted from Mori et al. 2017b)	334
Fig. 18.10	Right plot shows the analysis of the position error between the WR precipitation map (shaded background) and the SAR one (foreground) for the case study of Fig. 18.8. Values have been normalized to the maximum of the dataset. Note that the WR data precedes the SAR data by ~12 min. Left plot shows Complementary Cumulative Distribution Function (CCDF) for the same case study. Blue lines represent SAR data degraded at WR resolution (1000 m); red lines represent WR data. (Adapted from Mori et al. 2017b)	335
Fig. 19.1	PMW sensor Equator-crossing times for 12-24 Local Time (LT; 0000-1200 LT is the same) for the modern PMW sensor era. These are all ascending passes, except F08 is descending. Shading indicates that the precessing TRMM, Megha-Tropiques, and GPM cover all times of day with changes that are too rapid to depict at this scale. (Image by Eric Nelkin (SSAI; GSFC), 12 July 2018; https://pmm.nasa.gov/sites/default/files/imce/times_allsat.jpg holds the current version, last accessed 1 Apr. 2019)	345
Fig. 19.2	Rainfall accumulations for the week of 25-31 August 2017 over the US Gulf Coast for NOAA Multi-Radar Multi-Sensor (MRMS) data (left) and IMERG V05 Late estimates (right). Houston, Texas is just west of Area 1	350
Fig. 19.3	Time series of area-average rainfall for the week of 25-31 August 2017 over the US Gulf Coast for the near-coastal Area 1 (top) and the more inland Area 2 (bottom). Houston, Texas is just west of Area 1. The IMERG Late averages are labeled precipitationCal, and the IR-based precipitation time series is IRprecipitation	351
Fig. 20.1	Image of “JAXA Global Rainfall Watch” website (http://sharaku.eorc.jaxa.jp/GSMaP/ , last accessed 15 Oct. 2018)	357
Fig. 20.2	Process flowchart for the GSMaP product	359
Fig. 20.3	Distribution of the daily rain amount around Japan on June 25, 2017 for (a) JMA radar-AMeDAS, (b) GC V6, (c) MVK V6, (d) NRT V6, (e) NOW V6, and (f) Hydro-Estimator (H-E)	365

Fig. 20.4	Time series of the correlation coefficient (CC) for the GSMaP products and the H-E with reference to the JMA's Radar-AMeDAS around Japan in 0.25 deg lat/lon grid and daily accumulation from Apr. 1, 2017 to Dec. 31, 2018. Monthly mean was applied to the daily CC values. Open circles denote the GA; closed circles denote the MVK; plusses denote the NRT; crosses denote the NOW; and triangles denote the H-E	365
Fig. 20.5	Time series of the root mean square error (RMSE) for the GSMaP products and the H-E with reference to the JMA's Radar-AMeDAS around Japan in 0.25 deg lat/lon grid and daily accumulation from Apr. 1, 2017 to Dec. 31, 2018. Monthly mean was applied to the daily RMSE values. Open circles denote the GA; closed circles denote the MVK; plusses denote the NRT; crosses denote the NOW; and triangles denote the H-E	366
Fig. 20.6	Accumulated rainfall (mm year^{-1}) over 3 months: June to August 2015 for (a) MRMS, (b) MVK V6, and (c) GA V6. The MRMS was analyzed over areas where the RQI was more than 50	367
Fig. 20.7	Two-dimensional distribution function of daily precipitation over 3 months from June to August 2015. Horizontal axis shows the rain rate of the MRMS, and vertical axis shows rain rates of (a) MVK and (b) GA. Hatch color shows the sample number of occurrences. The average value (squares) and the standard deviation (1 sigma, bars) are shown for bins of 0.2 mm h^{-1} of the MRMS with the horizontal axis as a reference. The broken line is a one-to-one line	368
Fig. 21.1	The major processing modules and data flows in IMERG. The blocks are organized by contributing institution; the final code package is an integrated system. (Adapted from Huffman et al. 2018)	378
Fig. 21.2	Cloud image segmentation, feature extraction, classification, and rainfall estimation of the PERSIANN-CCS algorithm	379
Fig. 21.3	(a) The Tb-R relationship of 20×20 SOFM cloud patch groups and (b) the Tb-R curves with respect to the SOFM groups G0~G6	379
Fig. 21.4	PMW sensor Equator-crossing time for 1200–2400 local time for the modern PMW sensor era. Image by Eric Nelkin (SSAII GSFC, 5 July 2017; current version at https://precip.gsfc.nasa.gov/times_allsat.jpg , last accessed 5 Apr. 2019; also see Huffman et al. 2018)	380
Fig. 21.5	Matching rainfall from PMW satellites with the rainfall from PERSIANN-CCS using PMM rainfall. (Adapted from Karbalaee et al. 2017)	381

Fig. 21.6	(a) PERSIANN-CCS rainfall, (b) PMW recalibrated PERSIANN rainfall and (c) PMW rainfall maps at 0330 UTC, 9 March 2012. (rain rate: mm h^{-1})	382
Fig. 21.7	(a) cloud image from longwave infrared channel; (b) segmentation of IR cloud image at $T_b = 253 \text{ K}$, (c) segmentation of cloud image at $T_b = 260 \text{ K}$, and (d) segmentation of cloud image at $T_b = 280 \text{ K}$	385
Fig. 21.8	PERSIANN-CCS mapping curves using IR T_b thresholds at (a) 253 and (b) 280 K	386
Fig. 21.9	Rainfall map at 1600 UTC time 3 July 2012, from (a) Q2 radar map, (b) satellite image of IR brightness temperature, and the rainfall maps estimated from segmentation using thresholds from $T_b =$ (c) 253, (d) 260, and (e) 280 K	386
Fig. 22.1	The WMO World Weather Watch global distribution of the Regional Basis Synoptic Network weather stations providing SYNOP (surface synoptic observations) reports during October 2002. (WMO 2003). The colour denotes each station's reporting rate	394
Fig. 22.2	Schematic summarising the TAMSAT calibration and rainfall estimation process. Squares denote inputs or outputs and ovals denote processes. (Adapted from Maidment et al. 2017)	397
Fig. 22.3	TAMSAT v3.0 rainfall total (left) and rainfall anomaly (right) for August 2017	398
Fig. 22.4	Full disc (background) Meteosat-8 thermal-infrared image captured at 1800 UTC on 12 August 2005. The grey scale assigned signifies warmer (colder) surfaces as the darker (whiter) regions. A subset (foreground) of this image projected onto a longitude-latitude grid over West Africa giving the brightness temperature (scale in K) of the scene observed	399
Fig. 22.5	Schematic illustrating the assumptions in the TAMSAT algorithm. Clouds with tops colder than the threshold temperature (T_t) are assumed to be raining, while clouds with tops warmer than T_t are assumed not to be raining (schematic by David Grimes)	399
Fig. 22.6	TAMSAT v3.0 threshold temperature (T_t) maps for January (left) and August (right) respectively	401
Fig. 23.1	The number of monthly station data used in the production of the CHIRPS Final precipitation data	411
Fig. 23.2	CHIRPS2.1 schema	415
Fig. 23.3	Map of 14,197 stations used in the validation analysis	419
Fig. 23.4	Plots of station means (x-axis) and IRP (red diamonds), CHIRP2.0 (blue skinny diamonds) and CHIRP2.1 (yellow circles) means (y-axis). Data stratified binned by CHIRP2.0 values	420

Fig. 23.5	Empirical correction factors obtained from the station data and IRP averages (yellow circles) along with regression estimates based on the IRP means	420
Fig. 23.6	Change in standard deviation (CHIRPS2.1 minus 2.0). Red circles indicate a decline of -2 mm or more, orange circles indicate decreases of between -1 and -2 mm, grey circles indicate change values between -1 and 1 mm, cyan circles had changes between $+1$ and $+2$ mm, blue circles had changes greater than $+2$ mm	422
Fig. 23.7	Top – CHIRP POD for events with at least 20 mm of precipitation and change in $+20$ mm POD (CHIRPS2.1 minus 2.0). Middle – Map of changes in 20 + mm POD (CHIRPS2.1 minus 2.0). Cyan circles indicate POD increases between 5 and 10 percent, blue circles had changes greater than 10 percent. Average increase in POD was 11%. Bottom – Map of changes in FAR (CHIRPS2.1 minus 2.0). Orange circles indicate POD increases between 0 and 5 percent, red circles had changes greater than 5 percent. Average increase in FAR was 4%	424
Fig. 24.1	The relative uncertainty in % as a function of daily rain accumulation for land (left) and ocean (right) conditions for July 2012 all over the Tropics	435
Fig. 24.2	Flow chart of the merging algorithm	436
Fig. 24.3	Map of the July–September 2012–2016 average daily precipitation from TAPEER 1.5	438
Fig. 24.4	Hydrological application of TAPEER 1.5 over the Niger River basin. The left figure shows the total rainfall accumulated over the Niger river basin in 2016 (1 January–31 December), the right plot illustrates the simulation of the discharge in Niamey based on the hydrological model MGB (Fleischmann et al. 2018) with TAPEER rainfall as forcing	439
Fig. 24.5	Map of the monthly precipitation accumulation in mm for August 2016 from the full run of GIRAFE	442
Fig. 24.6	Map of the monthly mean difference between the two runs of GIRAFE with or without the sounders in the constellation (mm day^{-1})	443
Fig. 24.7	GPM imagers and sounders (in blue) and GPM imagers only (in red) precipitation volume distributions over all the region (top), over land (middle) and ocean (bottom)	444
Fig. 24.8	Zonal mean number of overpass of the full-blown microwave constellation including imagers, sounders and SAPHIR. Computations realized using the IXION package. (Capderou 2014)	446

List of Tables

Table 1.1	Satellites and sensors contributing to the GPM constellation	6
Table 3.1	MADRAS channel frequency and noise levels	49
Table 3.2	Noise and difference with simulations in K for each SAPHIR channels	50
Table 3.3	List of weather prediction agencies which are assimilating SAPHIR observations within an operational framework	57
Table 4.1	Typical MWI channels used for the retrieval of the physical variables $\odot$ and $\circ$ denotes necessary and important channels, respectively. S and L denotes over-sea and over-land retrievals, respectively	65
Table 4.2	Major satellite MWIs from the 1970s to the present	66
Table 4.3	Radiometric performance characteristics of ESMR on NIMBUS-5	67
Table 4.4	Radiometric performance characteristics of ESMR on NIMBUS-6	67
Table 4.5	Radiometric performance characteristics of SMMR on NIMBUS-7, SEASAT	67
Table 4.6	Radiometric performance characteristics of SSM/I on DMSP 5D satellites	68
Table 4.7	Radiometric performance characteristics of TMI on TRMM	69
Table 4.8	Radiometric performance characteristics of WindSat on Coriolis	70
Table 4.9	Radiometric performance characteristics of AMSR-E on Aqua .	71
Table 4.10	Radiometric performance characteristics of AMSR2 on GCOMW1	71
Table 4.11	Radiometric performance characteristics of GMI on GPM	71
Table 4.12	Major MWS satellite and instruments from the 1970's to present	72
Table 4.13	Radiometric performance of the MSU sensor on NOAA satellites	73

Table 4.14	Radiometric performance of the SSM/T sensor on DMSP satellites	74
Table 4.15	Radiometric performance of the SSMT/2 sensor on DMSP satellites	74
Table 4.16	Radiometric performance of the SSMIS sensor on the DMSP satellites	75
Table 4.17	Radiometric performance of the SAPHIR sensor on the M-T Satellite	77
Table 4.18	Radiometric performance of the AMSU-A sensor on NOAA and MetOp satellites	77
Table 4.19	Radiometric performance of the AMSU-B sensor on NOAA satellites	77
Table 4.20	Radiometric performance of the MHS sensor on NOAA satellites	78
Table 4.21	Radiometric performance of the ATMS sensor on the S-NPP and NOAA-20 satellites	79
Table 5.1	Payload complement of EPS and EPS-SG satellites	85
Table 5.2	Required MWI performance	86
Table 5.3	Required ICI performance	88
Table 5.4	CIMR channels complement and required performance according to current requirements (2018)	94
Table 7.1	Channel combinations and correlations against radar reflectivity (dBZ) (Alishouse et al. 1990)	129
Table 9.1	Summary of MiRS precipitation rate performance relative to Stage IV measurements over the CONUS for the period 1–30 March 2016	163
Table 9.2	Summary of MiRS precipitation rate performance relative to Stage IV measurements over the CONUS for the period 1–31 July 2016	163
Table 9.3	Categorical scores (Probability of Detection (%), Probability of False Detection (%), and Heidke Skill Score) of MiRS ATMS (Oper: operational version; Test: experimental test version) rain rates relative to Stage IV for a rain rate threshold of 0.5 mm h^{-1}	165
Table 9.4	Categorical scores (Probability of Detection (%), Probability of False Detection (%), and Heidke Skill Score) of MiRS ATMS (Oper: operational version; Test: experimental test version) rain rates relative to Stage IV for a rain rate threshold of 3.0 mm h^{-1}	166
Table 12.1	Comparisons of melting layer boundaries between different criteria for NAMMA, GRIP and Wakasa Bay data	198

Table 12.2	Comparison of stratiform, convective and other rain types between dual-frequency classification method and Ku only method	199
Table 12.3	Count of melting layer top match between dual-frequency method of DPR and Ku only method using GPM DPR data from tropical storms including cyclones, typhoons and hurricanes	200
Table 12.4	Count of melting layer bottom match between dual-frequency method and Ku only method using GPM DPR data from tropical storms including cyclones, typhoons and hurricanes	201
Table 12.5	Information on snow validation cases	207
Table 12.6	Meaning of abbreviation used for ground radar hydrometeor identification	207
Table 12.7	Average match ratio for validation cases under different surface types	208
Table 15.1	Databases of scattering properties for snow particles at microwave frequencies	258
Table 17.1	S-NPP SFR metrics before and after calibration	307
Table 17.2	S-NPP SD Metrics from satellite-only and hybrid algorithms ...	309
Table 17.3	S-NPP SFR validation metrics	310
Table 17.4	Metrics of S-NPP SFR 3-month (Jan–Mar 2017) average	311
Table 19.1	Lists of data field variable names and definitions to be included in each of the output datasets. Primary fields for users are in italics	348
Table 20.1	GSMaP product list in the GPM era	357
Table 20.2	Brief summary of evolutions in the algorithms from V6 to V7	363
Table 20.3	Values of the CC and RMSE averaged during 21 months: April 2017 to December 2018	366
Table 21.1	Statistical parameters used for global validation calculated for 8 zones (December, January, and February 2012). Statistics is provided based on the concurrent samples of test data over the 3-month period. Bias & corr (no unit); RMSE (mm/3-month) . .	383
Table 21.2	Statistical parameters used for global validation calculated for 8 zones (June, July, and August 2012). Statistics is provided based on the concurrent samples of test data over the 3-month period. Bias & corr (no unit); RMSE (mm/3-month)	383
Table 21.3	Comparison between PERSIANN-CCS and MA-PERSIANN-CCS with PMW satellite data over CONUS during winter and summer 2012. Statistics is provided based on the concurrent samples of test data over the 3-month period. Bias & corr (no unit); RMSE (mm/3-month)	384

Table 21.4	Comparison between PERSIANN-CCS, MA-PERSIANN-CCS, and PMW with Q2 ground based radar over CONUS during winter and summer 2012. Statistics is provided based on the concurrent samples of test data over the 3-month period. Bias & corr (no unit); RMSE (mm/3-month)	384
Table 22.1	Summary of widely used satellite rainfall datasets providing coverage for Africa	396
Table 22.2	Contingency table for determining T_t . The occurrence threshold is set at zero for both rainfall (mm) and CCD (hours). All gauge-CCD pairs are split into one of four possible groups ($n_{11}; n_{22}$) with the counts for each group recorded	400
Table 23.1	Counts of CHC station climate normals added to the GPCC and FAO archive	413
Table 23.2	Mean Bias Error (MBE) and Mean Absolute Error (MAE) by precipitation stratification. All statistics in mm per pentad	420
Table 23.3	Pentad validation statistics for events when stations observed at least 1 mm of rainfall	422
Table 24.1	The configuration of the geostationary infrared fleet for August 2016 used in the precipitation products	437
Table 24.2	The configuration of the passive microwave constellation (imagers and sounders) used for the various TAPEER and GIRAFE products	438

Contributors

Christophe Accadia European Organization for the Exploitation of Meteorological Satellites (EUMETSAT), Darmstadt, Germany

Robert F. Adler Earth System Science Interdisciplinary Center (ESSIC), University of Maryland, College Park, MD, USA

Kazumasa Aonashi Meteorological Research Institute (MRI), Japan Meteorological Agency (JMA), Tsukuba, Japan

Yoriko Arai Remote Sensing Technology Center of Japan (RESTEC), Tokyo, Japan

Moshe Armon The Fredy & Nadine Herrmann Institute of Earth Sciences, Hebrew University of Jerusalem, Jerusalem, Israel

Kristi Arsenault SAIC, Inc., McLean, VA, and National Aeronautics and Space Administration (NASA), Goddard Space Flight Center (GSFC), Greenbelt, MD, USA

Malarvizhi Arulraj Pratt School of Engineering, Civil & Environmental Engineering, Duke University, Durham, NC, USA

Hamed Ashouri Department of Civil and Environmental Engineering, Center for Hydrometeorology and Remote Sensing (CHRS), University of California, Irvine, CA, USA

Luca Baldini Institute of Atmospheric Sciences and Climate (ISAC), National Research Council (CNR), Roma, Italy

Ana P. Barros Pratt School of Engineering, Civil & Environmental Engineering, Duke University, Durham, NC, USA

Alessandro Battaglia Department of Physics and Astronomy, University of Leicester, Leicester, UK

Hylke E. Beck Department of Civil and Environmental Engineering, Princeton University, Princeton, NJ, USA

Andreas Becker Global Precipitation Climatology Center (GPCC), Deutscher Wetterdienst (DWD), Offenbach, Germany

Ali Behrangi Department of Hydrology and Atmospheric Sciences, University of Arizona, Tucson, AZ, USA

Hugo Bellenger Laboratoire de Météorologie Dynamique/IPSL, CNRS, Sorbonne Université, École Normale Supérieure, École Polytechnique, Paris, France
Japan Agency for Marine-Earth Science and Technology (JAMSTEC), Yokosuka, Japan

Emily Berndt NASA, Marshall Space Flight Center (MSFC), Huntsville, AL, USA

Emily Black Department of Meteorology, University of Reading, Reading, UK

David T. Bolvin Science Systems and Applications, Inc., Lanham, MD, and NASA/GSFC, Greenbelt, MD, USA

Rogério Bonifacio World Food Programme, Vulnerability Assessment and Mapping Unit, Roma, Italy

Sid-Ahmed Boukabara NOAA/NESDIS/STAR, College Park, MD, USA

Dominique Bouniol Météo France, Centre National de Recherches Météorologiques (CNRM), Groupe de Modélisation et d'Assimilation pour la Prévision (GMAP), OBS, Toulouse, France

Jacqueline Boutin Sorbonne Université, CNRS, Institut de Recherche pour le Développement (IRD), Muséum National d'Histoire Naturelle (MNHN), Laboratoire d'Océanographie et du Climat, Expérimentations et Approches Numériques (LOCEAN), Paris, France

Dan Braithwaite Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

Scott A. Braun NASA/GSFC, Greenbelt, MD, USA

Luca Brocca Research Institute for Geo-Hydrological Protection (IRPI), National Research Council (CNR), Perugia, Italy

Jörg Burdanowitz Institute for Meteorology, University of Hamburg, Hamburg, Germany

Steven J. Burian Department of Civil and Environmental Engineering, University of Utah, Salt Lake City, UT, USA

Stefania Camici CNR/IRPI, Perugia, Italy

Michel Capderou CNRS/LMD, Palaiseau, France

Daniele Casella CNR/ISAC, Roma, Italy

Elsa Cattani CNR/ISAC, Bologna, Italy

Daniel Cecil NASA/MSFC, Huntsville, AL, USA

Philippe Chambon Météo France, CNRM/GMAP/OBS, Toulouse, France

Venkatachalam Chandrasekar Department of Electrical and Computer Engineering, Colorado State University, Ft. Collins, CO, USA

Luca Ciabatta CNR/IRPI, Perugia, Italy

Sophie Cloché CNRS/IPSL, Palaiseau, France

Paolo Colucci EUMETSAT, Darmstadt, Germany

Salvatore D'Addio European Space Agency (ESA), European Space Research and Technology Centre (ESTEC), Noordwijk, The Netherlands

Frank Davenport Climate Hazards Group (CHG), University of California, Santa Barbara, CA, USA

Michel Dejus Centre National d'Études Spatiales (CNES), Toulouse, France

Tufa Dinku International Research Institute for Climate and Society (IRI), The Earth Institute at Columbia University, Palisades, NY, USA

Craig Donlon ESA/ESTEC, Noordwijk, The Netherlands

Mohamed El Saadani Department of Civil Engineering, University of Louisiana Lafayette, Lafayette, LA, USA

Ralph R. Ferraro NOAA/NESDIS/STAR, College Park, MD, USA

Robert D. Field Department of Applied Physics and Applied Mathematics, Columbia University, and NASA Goddard Institute for Space Studies, New York, NY, USA

Paolo Filippucci CNR/IRPI, Perugia, Italy

Thomas Fiolleau CNRS, Laboratoire d'Études en Géophysique et Océanographie Spatiales (LEGOS), Toulouse, France

Efi Foufoula-Georgiou Department of Civil and Environmental Engineering and Department of Earth Science, University of California, Irvine, CA, USA

Kevin Fuell Earth System Science Center (ESSC), University of Alabama in Huntsville, Huntsville, AL, USA

Chris Funk United States Geological Survey (USGS), Earth Resources Observation and Science (EROS) Center, Sioux Falls, SD, and CHG, University of California, Santa Barbara, CA, USA

Kinji Furukawa Japan Aerospace Exploration Agency (JAXA), Tokyo, Japan

Kevin Garrett NOAA/NESDIS/STAR, College Park, MD, USA

Alan Geer European Centre for Medium-range Weather Forecasts (ECMWF), Reading, UK

Radoslaw Goska Iowa Institute of Hydraulic Research (IIHR) – Hydroscience & Engineering, University of Iowa, Iowa City, IA, USA

Marielle Gosset Geoscience Environnement, Toulouse, France

Christopher Grassotti Cooperative Institute for Satellite Earth System Studies (CISESS), ESSIC, University of Maryland, College Park, MD, USA

Helen Greatrex Department of Meteorology, University of Reading, Reading, UK

Mircea Grecu Morgan State University, Baltimore, MD, and NASA/GSFC, Greenbelt, MD, USA

Guojun Gu ESSIC, University of Maryland, College Park, MD, USA

Adrien Guérou CNRS/LEGOS, Toulouse, France

Clément Guilloteau Department of Civil and Environmental Engineering, University of California, Irvine, CA, USA

Ziad S. Haddad Jet Propulsion Laboratory (JPL), California Institute of Technology (Caltech), Pasadena, CA, USA

Sebastian Hahn Department of Geodesy and Geoinformation, Research Group Remote Sensing, TU Wien, Vienna, Austria

Atsushi Hamada Faculty of Sustainable Design, University of Toyama, Toyama, Japan

Laura Harrison CHG, University of California, Santa Barbara, CA, USA

Negin Hayathini Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

John M. Haynes Cooperative Institute for Research in the Atmosphere (CIRA), Colorado State University, Ft. Collins, CO, USA

Ryan Honeyager UCAR, College Park, MD, USA

Kuo-Lin Hsu Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

George J. Huffman NASA/GSFC, Greenbelt, MD, USA

Toshio Iguchi National Institute of Information and Communications Technology (NICT), Koganei, Japan

Flavio Iturbide-Sanchez NOAA/NESDIS/STAR, College Park, MD, USA

Menglin Jin Department of Atmospheric and Oceanic Science, University of Maryland, College Park, MD, USA

Bradford Johnson Department of Geography, University of Georgia, Athens, GA, USA

Erin Jones NOAA/NESDIS/STAR, College Park, MD, USA

Robert J. Joyce Innovim, Greenbelt, MD, and NOAA, National Weather Service (NWS), Climate Prediction Center (CPC), College Park, MD, USA

Rômulo A. Jucá Oliveira Geoscience Environnement, Toulouse, France

Misako Kachi Earth Observation Research Center (EORC), JAXA, Ibaraki, Japan

Eugenia Kalnay Department of Atmospheric and Oceanic Science, University of Maryland, College Park, MD, USA

Sarah Kapnick NOAA, Geophysical Fluid Dynamics Laboratory (GFDL), Princeton, NJ, USA

Negar Karbalee Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

Nozomi Kawamoto RESTEC, Tokyo, Japan

Masahiro Kazumori JMA, Tokyo, Japan

Christopher Kidd ESSIC, University of Maryland, College Park, MD, and NASA/GSFC, Greenbelt, MD, USA

Dalia B. Kirschbaum NASA/GSFC, Greenbelt, MD, USA

Pierre-Emmanuel Kirstetter School of Meteorology and School of Civil Engineering and Environmental Sciences and Advanced Radar Research Center, University of Oklahoma and NOAA/National Severe Storms Laboratory, Norman, OK, USA

Marian Klein Boulder Environmental Sciences and Technology, Boulder, CO, USA

Ulf Klein ESA/ESTEC, Noordwijk, The Netherlands

Christian Klepp Max Planck Institute for Meteorology, Hamburg, Germany

Stefan Kneifel Institute for Geophysics and Meteorology, University of Cologne, Cologne, Germany

Pavlos Kollias School of Marine and Atmospheric Sciences, Stony Brook University, Stony Brook, NY, USA

Cezar Kongoli University of Maryland, College Park, MD, USA

Shunji Kotsuki RIKEN Center for Computational Science, Kobe, Japan
Center for Environmental Remote Sensing, Chiba University, Chiba, Japan

Witold F. Krajewski IIHR-Hydroscience & Engineering, University of Iowa, Iowa City, IA, USA

Takuji Kubota EORC/JAXA, Ibaraki, Japan

Paul A. Kucera University Corporation for Atmospheric Research (UCAR), Boulder, CO, USA

Mark S. Kulie NOAA/NESDIS/STAR, Advanced Satellite Products Branch, Madison, WI, USA

Christian D. Kummerow Department of Atmospheric Science, Colorado State University, Ft. Collins, CO, USA

Frank Lafontaine Jacobs, Engineering Services and Science Capability Augmentation (ESSCA), NASA/MSFC, Huntsville, AL, USA

Martin Landsfeld CHG, University of California, Santa Barbara, CA, USA

Sante Laviola CNR/ISAC, Bologna, Italy

Minda Le Department of Electrical and Computer Engineering, Colorado State University, Ft. Collins, CO, USA

Matthew D. Lebsack JPL/Caltech, Pasadena, CA, USA

Tristan S. L'Ecuyer Department of Atmospheric and Oceanic Sciences, University of Wisconsin-Madison, Madison, WI, USA

Jussi Leinonen JPL/Caltech, Pasadena, CA, USA

Anita LeRoy ESSC, University of Alabama in Huntsville, Huntsville, AL, USA

Vincenzo Levizzani CNR/ISAC, Bologna, Italy

Guo-Yuan Lien Research and Development Center, Central Weather Bureau, Taipei, Taiwan

Chuntao Liu Texas A&M University, Department of Physical and Environmental Sciences, Corpus Christi, TX, USA

Guosheng Liu Department of Earth, Ocean and Atmospheric Science, Florida State University, Tallahassee, FL, USA

Quanhua Liu NOAA/NESDIS/STAR, College Park, MD, USA

Shuyan Liu CIRA, Colorado State University, Ft. Collins, CO, USA

Eric Maddy NOAA/NESDIS/STAR, College Park, MD, USA

Viviana Maggioni Sid and Reva Dewberry Department of Civil, Environmental, and Infrastructure Engineering, George Mason University, Fairfax, VA, USA

Ross Maidment Department of Meteorology, University of Reading, Reading, UK

Anna Cinzia Marra CNR/ISAC, Roma, Italy

Francesco Marra The Fredy & Nadine Herrmann Institute of Earth Sciences, Hebrew University of Jerusalem, Jerusalem, Israel, and CNR/ISAC, Bologna, Italy

Frank S. Marzano Department of Information Engineering, Electronics and Telecommunications (DIET), Sapienza University of Roma, Roma, Italy

Takeshi Masaki RESTEC, Tokyo, Japan

Bathobile Maseko South African Weather Service, Pretoria, South Africa

Christian Massari CNR/IRPI, Perugia, Italy

Vinia Mattioli EUMETSAT, Darmstadt, Germany

Kevin McGrath Jacobs/ESSCA, NASA/MSFC, Huntsville, AL, USA

Amy McNally ESSIC, University of Maryland, College Park, MD, and NASA/GSFC, Greenbelt, MD, USA

Tomoaki Mega Tokyo Metropolitan University, Tokyo, Japan

Huan Meng NOAA/NESDIS/STAR, College Park, MD, USA

Andrés Merino Department of Chemistry and Applied Physics, University of León, León, Spain

Lisa Milani University of Maryland, College Park, MD, and NASA/GSFC, Greenbelt, MD, USA

Takemasa Miyoshi RIKEN Center for Computational Science, Kobe, Japan

Andrew Molthan NASA/MSFC, Huntsville, AL, USA

Saverio Mori DIET, Sapienza University of Roma, Roma, Italy

Efrat Morin The Fredy & Nadine Herrmann Institute of Earth Sciences, Hebrew University of Jerusalem, Jerusalem, Israel

Kenji Nakamura Dokkyo University, Saitama, Japan

Eric J. Nelkin Science Systems and Applications, Inc., Lanham, MD, and NASA/GSFC, Greenbelt, MD, USA

Brian R. Nelson NOAA/NESDIS, National Centers for Environmental Information (NCEI), Asheville, NC, USA

Phu Nguyen Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

Riko Oki Earth Observation Research Center (EORC)/Japan Aerospace Exploration Agency (JAXA), Ibaraki, Japan

William S. Olson University of Maryland Baltimore County, Baltimore, MD, and NASA/GSFC, Greenbelt, MD, USA

Mohammed Ombadi Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

Davide Ori Institute for Geophysics and Meteorology, University of Cologne, Cologne, Germany

Shigenori Otsuka RIKEN Center for Computational Science, Kobe, Japan

Ming Pan Department of Civil and Environmental Engineering, Princeton University, Princeton, NJ, USA

Giulia Panegrossi CNR/ISAC, Roma, Italy

Florian Pappenberger ECMWF, Shinfield Park, Reading, UK

Salvatore Pascale Department of Earth System Science, Stanford University, Stanford, CA, USA

Diego Pedreros USGS/EROS, Sioux Falls, SD, USA

Walter A. Petersen NASA/MSFC, Huntsville, AL, USA

Pete Peterson CHG, University of California, Santa Barbara, CA, USA

Nazzareno Pierdicca DIET, Sapienza University of Roma, Roma, Italy

Federico Porcù Department of Physics and Astronomy, University of Bologna, Bologna, Italy

Olivier P. Prat Cooperative Institute for Satellite Earth System Studies (CISESS), North Carolina State University, Asheville, NC, USA

Alain Protat Bureau of Meteorology (BoM), Melbourne, VIC, Australia

Felipe Quintero IIHR-Hydroscience & Engineering, University of Iowa, Iowa City, IA, USA

David L. Randel Department of Atmospheric Science, Colorado State University, Ft. Collins, CO, USA

Gilles Reverdin Sorbonne Université, CNRS/IRD/MNHN/LOCEAN, Paris, France

Sarah Ringerud ESSIC, University of Maryland, College Park, MD, and NASA/GSFC, Greenbelt, MD, USA

Rémy Roca CNRS/LEGOS, Toulouse, France

Geun-Hyeok Ryu National Meteorological Satellite Center, Korea Meteorological Administration, Seoul, South Korea

Paolo Sanò CNR/ISAC, Roma, Italy

Mathew R. P. Sapiano Sapiano Statistical Services, Atlanta, GA, USA

Masaki Satoh Atmosphere and Ocean Research Institute (AORI), The University of Tokyo, Chiba, Japan

Peter Schlüssel EUMETSAT, Darmstadt, Germany

Udo Schneider GPCC/DWD, Offenbach, Germany

Marc Schröder DWD, Offenbach, Germany

J. Marshall Shepherd Department of Geography, University of Georgia, Athens, GA, USA

Shoichi Shige Division of Earth and Planetary Sciences, Graduate School of Science, Kyoto University, Kyoto, Japan

Shraddhanand Shukla CHG, University of California, Santa Barbara, CA, USA

Gail M. Skofronick-Jackson NASA, Headquarters, Science Mission Directorate, Washington, DC, USA

Mark A. Smalley JPL/Caltech, Pasadena, CA, USA

Matthew Smith Information Technology and Systems Center, University of Alabama in Huntsville, Huntsville, AL, USA

Byung-Ju Sohn School of Earth and Environmental Sciences, Seoul National University, Seoul, South Korea

Hwan-Jin Song National Institute of Meteorological Sciences, Korea Meteorological Administration, Seoul, South Korea

Soroosh Sorooshian Department of Civil and Environmental Engineering/CHRS, University of California, Irvine, CA, USA

Thomas Stanley Universities Space Research Association (USRA), Columbia, MD, and NASA/GSFC, Greenbelt, MD, USA

Erich F. Stocker NASA/GSFC, Greenbelt, MD, USA

Alexandre Supply Sorbonne Université, CNRS/IRD/MNHN/LOCEAN, Paris, France

Nobuhiro Takahashi Institute for Space-Earth Environmental Research, Nagoya University, Nagoya, Japan

Yukari N. Takayabu Atmosphere and Ocean Research Institute, The University of Tokyo, Tokyo, Japan

Jackson Tan USRA, Columbia, MD, and NASA/GSFC, Greenbelt, MD, USA

Simone Tanelli JPL/Caltech, Pasadena, CA, USA

Wei-Kuo Tao NASA/GSFC, Greenbelt, MD, USA

Francisco J. Tapiador University of Castilla-La Mancha, Toledo, Spain

Elena Tarnavsky Department of Meteorology, University of Reading, Reading, UK

Tomoko Tashima EORC/JAXA, Ibaraki, Japan

Koji Terasaki RIKEN Center for Computational Science, Kobe, Japan

Ali Tokay University of Maryland Baltimore County, Baltimore, MD, and NASA/GSFC, Greenbelt, MD, USA

Hirofumi Tomita RIKEN Center for Computational Science, Kobe, Japan

Frederic Tridon Earth Observation Science, Department of Physics and Astronomy, University of Leicester, Leicester, UK

F. Joseph Turk JPL/Caltech, Pasadena, CA, USA

Jani Tyynelä Finnish Meteorological Institute (FMI), Helsinki, Finland

Tomoo Ushio Tokyo Metropolitan University, Tokyo, Japan

Albert I. J. M. van Dijk Fenner School of Environment & Society, The Australian National University, Canberra, Australia

Jean-Luc Vergely ACRI-st, Guyancourt, France

Noemi Vergopolan Department of Civil and Environmental Engineering, Princeton University, Princeton, NJ, USA

Daniel Vila Instituto Nacional de Pesquisas Espaciais (IPE), Centro de Previsão de Tempo e Estudos Climáticos (CPTEC), Cachoeira Paulista, Brazil

Wolfgang Wagner Department of Geodesy and Geoinformation, Research Group Remote Sensing, TU Wien, Vienna, Austria

Jian-Jian Wang ESSIC, University of Maryland, College Park, MD, USA

Jianxin Wang Science Systems Applications International and NASA/GSFC, Greenbelt, MD, USA

Graham P. Weedon Joint Centre for Hydro-Meteorological Research, Met Office, Wallingford, UK

Tobias Wehr ESA/ESTEC, Noordwijk, The Netherlands

David B. Wolff NASA/GSFC, Wallops Flight Facility, Wallops Island, VA, USA

Eric F. Wood Department of Civil and Environmental Engineering, Princeton University, Princeton, NJ, USA

Norman B. Wood Space Science and Engineering Center (SSEC), University of Wisconsin-Madison, Madison, WI, USA

Pingping Xie NOAA/NWS/CPC, College Park, MD, USA

Moeka Yamaji EORC/JAXA, Ibaraki, Japan

Munehisa K. Yamamoto Division of Earth and Planetary Sciences, Graduate School of Science, Kyoto University, Kyoto, Japan

Hisashi Yashiro Satellite Observation Center, National Institute for Environmental Studies, Tsukuba, Japan

Matthew Young Department of Meteorology, University of Reading, Reading, UK

Bradley Zavodsky NASA/MSFC, Huntsville, AL, USA

Acronyms

ABI	Advanced Baseline Imager (GOES)
ACE	Aerosol-Clouds-Ecosystem Mission (NASA)
AD	Analog-to-Digital converter
ADDA	Amsterdam DDA
AET	Actual Evapotranspiration
AGL	Above Ground Level
AHPS	Advanced Hydrologic Prediction Service (NWS)
AI	Aridity Index
AIP	Algorithm Intercomparison Programme
Air-MSPI	Airborne Multi-angle SpectroPolarimetric Imager (NASA)
AIRS	Atmospheric Infrared Sounder (NASA)
AKDT	Alaska Daylight Time
ALEXI	Atmosphere-Land Exchange Inverse
AMeDAS	Automated Meteorological Data Acquisition System (JMA)
AMIE/DYNAMO	ARM Madden-Julian Oscillation Investigation Experiment/ Dynamics of the Madden-Julian Oscillation
AMIP	Atmospheric Model Intercomparison Project (WCRP)
AMMA	African Monsoon Multidisciplinary Analysis experiment
AMO	Atlantic Meridional Oscillation
AMPR	Advanced Microwave Precipitation Radiometer (NASA)
AMS	Annual Maximum Series (of rainfall)
AMSL	Above Mean Sea Level
AMSR	Advanced Microwave Scanning Radiometer (JAXA)
AMSR-E	AMSR-EOS (NASA)
AMSU	Advanced Microwave Sounding Unit (NOAA and EUMETSAT)
AMW	Active Microwave
APHRODITE	Asian Precipitation—Highly Resolved Observational Data Integration Towards Evaluation of Water Resources (Japan)

APR-3	Airborne Third Generation Precipitation Radar (NASA)
APSIM	Agricultural Production Systems sIMulator
AR	Atmospheric River
ARC (1)	Africa Rainfall Climatology (NOAA)
ARC (2)	Active Radar Calibrator
ARM	Atmospheric Radiation Measurement (DoE)
ARM-SGP	ARM Southern Great Plains (DoE)
ARMAR	Airborne Rain-Mapping Radar (NASA and JPL)
ASCAT	Advanced SCATterometer (ESA)
ASCII	American Standard Code for Information Interchange
ASI	Italian Space Agency
ASL	Above Sea Level (a.s.l.)
ASTRAIA	Analyse Stereoscopique par Radar Aeroporte (CNRS)
ATBD	Algorithm Theoretical Basis Document
ATMS	Advanced Technology Microwave Sounder (NASA/NOAA)
AWARE	ARM West Antarctic Radiation Experiment
AWIPS	Advanced Weather Interactive Processing System (NWS and UNIDATA)
BAECC	Biogenic Aerosols-Effects on Clouds and Climate Experiment (ARM)
BB	Bright Band
BC	British Columbia
BCS	Bias Correction Scheme
BMKG	Badan Meteorologi, Klimatologi, dan Geofisika (Indonesia)
BoM	Bureau of Meteorology (Australia)
BRAIN	Bayesian Rain Algorithm Including Neural Networks (Megha-Tropiques)
BSA	Backscatter Alignment
BUFR	Binary Universal Form for the Representation of Meteorological Data
BUI	Buildup Index (FWI)
CAPE	Convective Available Potential Energy
CAPRICORN	Clouds, Aerosols, Precipitation, Radiation, and Atmospheric Composition over the Southern Ocean
CARE	Centre for Atmospheric Research Experiments (Environment Canada)
CATDS	Centre Aval de Traitement des Données SMOS
CC	Correlation Coefficient
CCD	Cold Cloud Duration
CCDF	Complementary Cumulative Distribution Function
CCI	Climate Change Initiative (ESA)
CCI	Commission for Climatology (WMO)
CCP (1)	Clouds, Convection, and Precipitation
CCP (2)	Cloud and Precipitation Process Mission

CCZ	Continent-Climate Zone
CDD	Consecutive Dry Days Index (ETCCDI)
CDF	Cumulative Density Function
CDR	Climate Data Record
CDRD	Cloud Dynamics and Radiation Database
CEMADEN	Centro Nacional de Monitoramento e Alertas de Desastres Naturais (Brazil)
CEOS	Committee on Earth Observation Satellites
CERES	Clouds and the Earth's Radiant Energy System (NASA)
CESM	Community Earth System Model
CFAD	Contoured Frequency by Altitude Diagram
CG	Cloud-to-Ground Lightning
CGMS	Coordination Group for Meteorological Satellites
CHC	Climate Hazards Center (University of California, Santa Barbara)
CHIRPS	Climate Hazards center InfraRed Precipitation with Stations
CHPcli	Climate Hazards Group's Precipitation Climatology
CIMR	Copernicus Imaging Microwave Radiometer Mission (EU)
CIndO	Central Equatorial Indian Ocean index
CIRA	Cooperative Institute for Research in the Atmosphere (CSU)
CLC	Corine Land Cover
CLIVAR	Climate Variability and Predictability (WMO)
CLW	Cloud Liquid Water
CLWC	Cloud Liquid Water Content
CMA	China Meteorological Administration
CMAP	CPC Merged Analysis of Precipitation
CMIP	Coupled Model Intercomparison Project (WCRP)
CMIP-5	CMIP Phase 5
CMORPH	CPC MORPHing algorithm
CMORPH-CRT	CMORPH Bias Corrected
CMORPH-KF	CMORPH-Kalman Filter
CM-SAF	Climate Monitoring-SAF (EUMETSAT)
CNES	Centre National D'Études Spatiales (France)
CNRM/GAME	Centre National de Recherches Météorologiques—Groupe d'études de l'Atmosphère Météorologique (Météo France)
CNR	Consiglio Nazionale delle Ricerche (Italy)
CNR-IRPI	CNR-Istituto di Ricerca per la Protezione Idrogeologica
CNR-ISAC	CNR-Istituto di Scienze dell'Atmosfera e del Clima
CNRS	Centre National de la Recherche Scientifique (France)
CNTL	Control Run
COADS	Comprehensive Ocean Atmosphere Data Set
ConQ	Moisture Flux Convergence
CONUS	Conterminous US
CORRA	Combined Radar-Radiometer Product (GPM)

CoSMIR	Conical Scanning Millimeter-wave Imaging Radiometer (NASA)
CPC	Climate Prediction Center (NOAA)
CPI	Convective Percent Index
CPL	Cloud Physics Lidar (NASA)
CPR	Cloud Profiling Radar (NASA)
CRS	Cloud Remote Sensing Radar (NASA)
CrIS	Cross-Track Infrared Sounder (NASA)
CRM	Cloud-Resolving Model
CRS	Cloud Radar System (NASA)
CRTM	Community Radiative Transfer Model
CRU	Climate Research Unit (Univ. of East Anglia)
CSA	Climate Service for Agriculture (Rwanda)
CSH	Convective and Stratiform Heating
CSI	Critical Success Index
CSK	COSMO-SkyMed (ASI)
CSP	Climate Services Partnership
CSU	Colorado State University
CSPP	Community Satellite Processing Package
CT	Cloud Thickness
CTH	Cloud Top Height
CWD	Consecutive Wet Days (ETCCDI)
CWV	Columnar Water Vapor
CYGNSS	Cyclone Global Navigation Satellite System (NASA)
DA	Data Assimilation
DAR	Differential Absorption Radar
DB	Dark Band
DBNet	Direct Broadcast Network
DC	Drought Code (FWI)
DD	Downward Decreasing
DDA	Discrete Dipole Approximation
DDSCAT	Discrete Dipole Scattering
DEM	Digital Elevation Model
DFR	Dual-Frequency Ratio
DI	Downward Increasing
DJF	December-January-February
DKRZ	Deutsches Klimarechenzentrum (Germany)
DLR	Deutsches Zentrum für Luft- und Raumfahrt (Germany)
DMC	Duff Moisture Code (FWI)
DMIP2	Distributed Hydrologic Model Intercomparison Project–Phase 2 (NWS)
DMSP	Defense Meteorological Satellite Program (US Navy)
DNN	Deep Neural Network

DoE	Department of Energy
DO-Op	Daylight Only Operations (CloudSat)
DoW	Doppler on Wheels (Center for Severe Weather Research)
DP	Dual Polarimetric Radar
DPC (1)	Data Processing Center (CloudSat)
DPC (2)	Department of Civil Protection of Italy
DPCA	Displaced Phase Center Antenna
DPR	Dual-frequency Precipitation Radar (GPM)
DRC	Democratic Republic of the Congo
DryMOD	Dryland hydrological MODel (University of Reading)
DSD	Drop Size Distribution
DSI	Drought Severity Index
DSSAT	Decision Support System for Agrotechnology Transfer
DWR	Dual-Wavelength Ratio
DYNAMO	Dynamics of the MJO experiment
D3R	Dual-Frequency Dual-Polarized Doppler Radar (NASA)
EA	East Africa
EAF	East Africa
EarthCARE	Earth Clouds, Aerosols, and Radiation Explorer (ESA-JAXA)
EASE	Equal-Area Scalable Earth
EBCM	Extended Boundary Condition Method
EC	European Commission
ECCC	Environment and Climate Change Canada
ECDI	Enhanced Combined Drought Index
ECMWF	European Centre for Medium-Range Weather Forecasts
EDF	Environmental Data Fusion
EDOP	ER-2 Doppler Radar
EDR	Environmental Data Record
EEA	Eastern East Africa
EIA	Earth Incident Angle
ELDORA	Electra Doppler Radar (NCAR)
EM	Electromagnetic
EMA	Effective Medium Approximations
eMAs	extended MODIS Airborne Simulator (NASA)
EMSR	Electronically Scanning Microwave Radiometer (NOAA)
ENACTS	Enhancing National Climate Services
ENSO	El Niño Southern Oscillation
EOF	Empirical Orthogonal Function
EOS	Earth Observing System (NASA)
EPD2	Extreme Precipitation Day > 2 in day ⁻¹
EPD4	Extreme Precipitation Day > 4 in day ⁻¹
EPS	EUMETSAT Polar System
EPSAT-SG	Estimation of Precipitation by Satellite Second Generation (CNRS-LMD)

EPS-SG	EPS Second Generation
ERA	ECMWF Reanalysis
ESA	European Space Agency
ESA-CCI	ESA Climate Change Initiative
ESMR	Electronically Scanned Microwave Radiometer (NOAA)
ESPC	Environmental Satellite Processing Center (NESDIS)
ESSIC	Earth System Science Interdisciplinary Center
ET	Evapotranspiration
ETCCDI	Expert Team on Climate Change Detection and Indices
EU	European Union
EUMETCast	EUMETSAT's Multicast Distribution System
EUMETSAT	European Organization for the Exploitation of Meteorological Satellites
EVI-3	Earth Venture Instrument-3 program (NASA)
EVT	Extreme Value Theory
EWFN	Energy-Water-Food Nexus
EWS	Early Warning System
EXRAD	ER-2 X-band Radar (NASA)
FAO	Food and Agriculture Organization (UN)
FAR	False Alarm Rate
FAS	Foreign Agricultural Service (USDA)
FB	Frequency Bias
FCDR	Fundamental Climate Data Record
FD	Frost Days
FDRS	Fire Danger Rating Systems
FEWS NET	Famine Early Warning Systems Network
FFMC	Fine Fuel Moisture Code (FWI)
FL	Freezing Level
FLDAS	FEWS NET Land Data Assimilation System
FLOR	Forecast-oriented Low Ocean Resolution model (GFDL)
FMI	Finnish Meteorological Institute
FNMOC	Fleet Numerical Meteorology and Oceanography Center (US Navy)
FoV	Field of View
FP	Forward Processing (GMAO)
FRMSE	Fractional RMSE
FSOI	Forecast Sensitivity Observation Impact
FWI	Fire Weather Index
GAGES	Geospatial Attributes of Gages for Evaluating Streamflow (USGS)
GANAL	Global Analysis (JMA)
GATE	Global Atmospheric Research Program Atlantic Tropical Experiment
GC	Ground Clutter

GCEM	Goddard Cumulus Ensemble Model (NASA)
GCM	General Circulation Model
GCOM	Global Change Observation Mission (JAXA)
GCOM-W	GCOM-Water
GCOS	Global Climate Observing System (WMO)
GCPE _x	GPM Cold Season Precipitation Experiment
GDAP	GEWEX Data and Assessment Panel
GDAS/GFS	Global Data Assimilation System/Global Forecast System (NOAA)
GEC	Geocoded Ellipsoid Corrected
GEO (1)	Geostationary orbit
GEO (2)	Group on Earth Observations
GEOGLAM	GEO Global Agricultural Monitoring
GEOS	Goddard Earth Observing System (NASA)
GEOS FP	Global Earth Observing System Forward Processing
GEOSS	Group on Earth Observation System of Systems
GeoSTAR	Geostationary Synthetic Thinned Aperture Radiometer (NASA)
GES DISC	Goddard Earth Sciences Data and Information Services Center
GEV	Generalized Extreme Value Distribution
GEWEX	Global Energy and Water Exchanges (WCRP)
GFCS	Global Framework for Climate Services (WMO)
GFDL	Geophysical Fluid Dynamics Laboratory (NOAA and Princeton University)
GFS	Global Forecast System (NCEP)
GFWED	Global Fire Weather Database (NASA GISS)
GHA	Greater Horn of Africa
GHCN-D	Global Historical Climatology Network-Daily (NOAA)
GHM	Global Hydrological Model
GHRC	Global Hydrology Resource Center (NASA)
GIRAFE	Global Interpolated RAINFall Estimation
GIS	Geographic Information System
GISS	Goddard Institute for Space Studies (NASA)
GLC	Global Landslide Catalog (NASA)
GMa	Gulf of Mexico area
GMAO	Global Modeling and Assimilation Office (NASA)
GMI	GPM Microwave Imager
GMM	Generalized Multiparticle Mie-solution
GoAMAZON	Green Ocean Amazon Experiment
GOES	Geostationary Operational Environmental Satellite (NOAA)
GOSAT-3	Greenhouse Gases Observing Satellite (JAXA)
GOTM	General Ocean Turbulence Model
GPC	Global Precipitation Climatology Center (DWD)
GPCP	Global Precipitation Climatology Project (GEWEX)

GPI	GOES Precipitation Index
GPM	Global Precipitation Measurement mission (NASA and JAXA)
GPM-CO	GPM Core Observatory
GPP	Gross Primary Production
GPROF	Goddard Profiling Algorithm (NASA)
GPS	Global Positioning System
GRACE	Gravity Recovery and Climate Experiment (NASA and DLR)
GRACE-FO	GRACE Follow-On
GRDC	Global Runoff Data Centre
Grid-Sat	Gridded Satellite Data (NOAA)
GRIP	Genesis and Rapid Intensification Processes (NASA)
GRISO	Rainfall Generator of Spatial Interpolation from Observation
GS	Grain Size
GSFC	Goddard Space Flight Center (NASA)
GSI	Gridpoint Statistical Interpolation
GSM	Global Spectral Model
GSMaP	Global Satellite Mapping of Precipitation (Japan)
GSMaP-MVK	GSMaP Motion Vector Kalman
GSMaP-NRT	GSMaP Near Real Time
GSOD	Global Surface Summary of the Day (NOAA)
GT	Gaussian Transformation
GTC	Geocoded Terrain Corrected
GTS	Global Telecommunication System
GV	Ground Validation
GW	Global Warming
GWIS	Global Wildfire Information System
GWP	Graupel Water Path
G5NR	GEOS-5 Nature Run (NASA)
HADS	Hydrometeorological Automated Data System (NWS)
HAMSR	High-Altitude MMIC Sounding Radiometer (JPL)
HDF	Hierarchical Data Format
HEMT	High Electron Mobility Transistor
HEPEX	Hydrological Ensemble Prediction Experiment
HID	Hydrometeor Identification algorithms
HIRS	High-Resolution Infrared Radiation Sounder (NOAA)
HISA	Hurricane Intensity and Structure Algorithm
HIWRAP	High-Altitude Imaging Wind and Rain Airborne Profiler (NASA)
HLM	Hillslope-Link Model
HMA	High Mountain Asia
HOAPS	Hamburg Ocean Atmosphere Parameters and Fluxes from Satellite Data (University of Hamburg)
HQPrecip	High-Quality Precipitation (GPM)

HR	Hit Rate
HRPP	High-Resolution Precipitation Product
HRWS	High-Resolution Wide Swath
H-SAF	Support to Operational Hydrology and Water Management (EUMETSAT)
HSS	Heidke Skill Score
HyMeX	Hydrological Cycle in Mediterranean Experiment
IC	Intra-cloud Lightning
ICDC	Integrated Climate Data Center (University of Hamburg)
ICE-POP	International Collaborative Experiment for the PyeongChang Olympics and Paralympics Experiment 2018
ICHARM	International Centre for Water Hazard and Risk Management
ICI	Ice Cloud Imager (EUMETSAT)
ICO-LETKF	Icosahedral LETKF
IDF	Intensity-Duration-Frequency Curve
IDW	Inverse Distance Weighting
IFAS	Integrated Flood Analysis System
IFM	Index Flood Method
IMERG	Integrated Multi-satellitE Retrievals for GPM
IMERG_E	IMERG Early Run (near real time with a latency of 6 h)
IMERG_F	IMERG Final Run (gauged-adjusted with a latency of 4 months)
IMERG_L	IMERG Late Run (reprocessed near real time with a latency of 18 h)
IOD	Indian Ocean Dipole
IPHEX	Integrated Precipitation and Hydrology Experiment (GPM)
IPS	Institut Pierre Simon Laplace (CNRS)
IPWG	International Precipitation Working Group (CGMS)
IR	Infrared
IRI	International Research Institute for Climate and Society (Columbia University)
IRP	Infrared Precipitation Estimate
ISI	Initial Spread Index (FWI)
ISRO	Indian Space Research Organisation
ISS	International Space Station
ITCZ	Intertropical Convergence Zone
IWP	Ice Water Path
I&Q	In-Phase and Quadrature Signal
JAXA	Japan Aerospace Exploration Agency
JCOMM	Joint Technical Commission for Oceanography and Marine Meteorology (WMO)
JCSDA	Joint Center for Satellite Data Assimilation (NOAA)
JERD	JPSS NESDIS ESPC Requirements Document
JF	January-February

JJA	June-July-August
JJAS	June-July-August-September
JMA	Japan Meteorological Agency
JPL	Jet Propulsion Laboratory
JPSS	Joint Polar Satellite System
JRA55	Japanese 55-year Reanalysis
JRC	Joint Research Centre (EC)
KGE	Kling-Gupta Efficiency
KMA	Korea Meteorological Administration
KWAJEX	Kwajalein Experiment
LACA&D	Latin American Climate Assessment & Dataset
LDM	Local Data Manager (UNIDATA)
LDR	Linear Depolarization Ratio
LEGOS	Laboratoire d'Études en Géophysique et Océanographie Spatiales
LEO	Low Earth Orbit
LETKF	Local Ensemble Transform Kalman Filter (RIKEN)
LFM	Local Forecast Model
LH	Latent Heating
LHASA	Landslide Hazard Assessment for Situational Awareness (NASA)
LIA	Local Incidence Angle
LLCF	Low-Level Clouds and Fog
LL-LETKF	Latitude-Longitude LETKF
LMD	Laboratoire de Météorologie Dynamique (CNRS)
LMODEL	Lagrangian Model (UC Irvine and University of Hull)
LPVEx	Light Precipitation Validation Experiment (GPM)
LR	Logistic Regression
LST	Land Surface Temperature
LUT	Lookup Table
LWP	Liquid Water Path
LZA	Local Zenith Angle
L1SR	Level 1 Science Requirements (GPM)
MADRAS	Microwave Analysis and Detection of Rain and Atmospheric Structures (Megha-Tropiques)
MAE	Mean Absolute Error
MAFF	Ministry of Agriculture, Forestry and Fisheries (Japan)
MAM	March-April-May
MARSOP	Monitoring Agricultural ResourceS Operational (JRC)
MBE	Mean Bias Error
MCS	Mesoscale Convective System
MCTA	Merged CloudSat, TRMM, and AMSR product
MC3E	Mid-latitude Continental Convective Clouds Experiment
ME	Mean Error

Medicane	Mediterranean hurricane
MERRA	Modern-Era Retrospective analysis for Research and Applications (NASA)
MGD	Multilook Ground Detected
MGDSST	Merged Satellite and In Situ Data Global Daily SST (JMA)
MHEMT	Metamorphic HEMT
MHOPrEx	Monsoon Himalaya Orographic Precipitation Experiment
MHS	Microwave Humidity Sounder (EUMETSAT)
MicroMAS-2	Micro-sized Microwave Atmospheric Satellite-2
MIIDAPS	Multi-instrument Inversion and Data Assimilation Preprocessing System (NOAA)
MIR	Middle IR
MIRA	Microwave/Infrared Rainfall Algorithm
MIRAS	Microwave Imaging Radiometer using Aperture Synthesis
MiRS	Microwave Integrated Retrieval System (NOAA)
MISDe	Modello Idrologico Semi-Distribuito in continuo (CNR-IRPI)
MLP	Melting Level Precipitation
MLS	Microwave Limb Sounder (EOS)
MMIC	Millimeter-Wave Monolithic Integrated Circuits
MODIS	Moderate Resolution Imaging Spectroradiometer (NASA)
MPE	Multisensor Precipitation Estimator
MP-M	Max-Planck-Institut für Meteorologie (Germany)
MRR	Micro Rain Radar
MREA	Modified Regression Empirical Algorithm
MRMS	Multi-Radar/Multisensor Precipitation Data
MREA	Modified Regression Empirical Algorithm
MRMS	Multi-Radar Multisensor system (NOAA)
MS	Multiple Scattering
MSG	Meteosat Second Generation (EUMETSAT)
MSG-CPP	MSG Cloud Physical Properties (EUMETSAT)
MSLP	Mean Sea-Level Pressure
MSPPS	Microwave Surface and Precipitation Products System (NOAA)
MSU	Microwave Sounding Unit (NOAA)
MSWEP	Multisource Weighted-Ensemble Precipitation
MTSAT	Multifunctional Transport Satellites (JMA)
MW	Microwave
MWCC	Microwave Cloud Classification
MWCOMB	Combined MW Rainfall Retrieval (CPC)
MWHS-2	Microwave Humidity Sounder-2 (CMA)
MWI	Microwave Imager (EUMETSAT)
MWR (1)	Microwave Radiometer
MWR (2)	Moving Window Regression

MWS	Microwave Sounder
NAa	North Atlantic Area
NAMMA	NASA African Monsoon Multidisciplinary Analyses
NAS	National Academies of Sciences (US)
NASA	National Aeronautics and Space Administration
NCA	North and Central America
NCAR	National Center for Atmospheric Research
NCE	National Centers for Environmental Information (NOAA)
NCEI	National Centers for Environmental Information
NCEP	National Centers for Environmental Prediction (NOAA)
NCEP-CFSR	Climate Forecast System Reanalysis
NCL	NCAR Command Language
NDVI	Normalized Difference Vegetation Index
NESDIS	National Environmental Satellite, Data, and Information Service (NOAA)
netCDF	Network Common Data Form
NEWS	NASA Energy and Water Cycle Study
NEXRAD	Next-Generation Weather Doppler Radar (NOAA/NWS)
NH	Northern Hemisphere
NHM	National Hydrometeorological Service
NICAM	Nonhydrostatic Icosahedral Atmospheric Model (RIKEN)
NIR	Near IR
NM	National Meteorology Agency
NMAE	Normalized Mean Absolute Error
NMHS	National Meteorological and Hydrological Service
NMQ	National Mosaic Quantitative Precipitation Estimation (NOAA/NSSL)
NU-WRF	NASA-Unified WRF
NPP	Net Primary Production
NRCS	Normalized Radar Cross Section
NOAA	National Oceanic and Atmospheric Administration
NOP	Numerical Ocean Prediction
NPOL	NASA S-Band Dual Polarimetric
NRL	Naval Research Laboratory (US Navy)
NRMSE	Normalized Root Mean Square Error
NRT	Near Real Time
NSE	Nash-Sutcliffe Efficiency
NSMC	National Satellite Meteorological Center (CMA)
NSSL	National Severe Storms Laboratory (NOAA)
NTPa	North Tropical Pacific Area
NUBF	Nonuniform Beam Filling
NU-WRF	NASA-Unified Weather Research and Forecasting
NWP	Numerical Weather Prediction
NWS	National Weather Service (NOAA)

OBCT	On-Board Calibration Target
OCE	Oceania
OceanRain	Ocean Rainfall And Ice-Phase Precipitation Measurement Network
OE	Optimal Estimation
OI	Optimal Interpolation
OLYMPEX	Olympic Mountains Experiment
OM	Observatoire Midi-Pyrénées
OND	October-November-December
OSCAR	Observing Systems Capability Analysis and Review (WMO)
OSPO	Office of Satellite and Product Operations (NOAA)
OSSE	Observing Systems Simulation Experiment
OZA	Observation Zenith Angle
PAW	Percentage Available Water
PCT	Polarization Corrected Temperature
PDF (1)	Probability Density Function
PDF (2)	Particle Distribution Function
PDO	Pacific Decadal Oscillation
PDSI	Palmer Drought Severity Index
PERHPP	Program for the Evaluation of High-Resolution Precipitation Products
PERSIANN	Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks (UC Irvine)
PERSIANN-CCS	PERSIANN-Cloud Classification System
PERSIANN-CDR	PERSIANN-Climate Data Records
PERSIANN-MSA	PERSIANN-Multispectral Analysis
PF	Precipitation Feature
PHENT	Pseudomorphic HEMT
PHIVOLCS	Philippine Institute of Volcanology and Seismology
PIA	Path-Integrated Attenuation
PICSA	Participatory Integrated Climate Services
PIP	Precipitation Imaging Package
PMA	Probability Matching Algorithm
PMI	Polarimetric Microwave Imager (CMA)
PMM (1)	Precipitation Measurement Mission (NASA)
PMM (2)	Probability Matching Method
PMP	Probable Maximum Precipitation
PMW	Passive Microwave
PNPR	Passive Microwave Neural Network Precipitation Retrieval (CNR-ISAC)
POD	Probability of Detection
POES	Polar Operational Environmental Satellites (NOAA)
POFD	Probability of False Detection
PoP	Probability of Precipitation

POS	Probability of Snowfall
PPI	Plan Position Indicator
PPP	Precipitation Per Person
PPS	Precipitation Processing System (IMERG)
PQPE	Probabilistic Quantitative Precipitation Estimation
PR	Precipitation Radar (TRMM)
PRCPTOT	Total Rainfall Amount (ETCCDI)
PRISM	Parameter-Elevation Regressions on Independent Slopes Model (Oregon State University)
PRPS	Precipitation Retrieval and Profiling Scheme
PSD	Particle Size Distribution
PSS	Practical Salinity Scale
PTH	Precipitation Top Height
PUSH	Precipitation Uncertainties for Satellite Hydrology
PV	Physical Validation (GPM)
PW	Precipitable Water
PWC	Precipitable Water Content
QA	Quality Assurance
QC	Quality Control
QCLCD	Quality Controlled Local Climatological Data
QI	Quality Index
QPE	Quantitative Precipitation Estimation
RADAP	Radar Data Processor (NOAA)
RADAR	Radio Detection and Ranging
RADEX	Radar Definition Experiment (OLYMPEX)
RCM	Regional Climate Model
RCS	Radar Cross Section
REA	Regressive Empirical Algorithm
REFAME	Rain Estimation Using Forward Adjusted-Advection of Microwave Estimates (UC Irvine)
RFC	River Forecast Center (NWS)
RFE	Rainfall Estimate (FAO)
RFI	Radio Frequency Interference
RGA	Rayleigh-Gans Approximation
RGB	Red Green Blue
RH	Relative Humidity
RHI	Range Height Indicator
RICO	Rain in Cumulus over the Ocean
RIM	Rain Impact Model (salinity)
RMS	Root Mean Square
RMSD	RMS Deviation
RMSE	RMS Error
RoF	Rain on Fog
RoFCC	RoF and Cap Clouds

ROI	Region of Interest
RoLLC	Rain on Low-Level Clouds
ROSA	Radio Occultation Sensor for Atmosphere
RQI	Radar Quality Index
RR	Rain Rate
RRFA-S	Regional Rainfall Frequency Analysis using Satellite Precipitation
RSS	Remote Sensing Systems Inc.
RTH	Rain Top Height
RTM	Radiative Transfer Model
RTTOV	Radiative Transfer for TOVS
RV (or (R/V)	Research Vessel
RWH	Rainwater Harvesting
RW	Rainwater Path
R2O/O2	Research-to-Operations/Operations-to-Research paradigm (SPoRT)
SAa	South Atlantic Area
SAF	Satellite Application Facility (EUMETSAT)
SAM (1)	System for Atmospheric Modeling
SAM (2)	Southern Appalachian Mountains
SAPHIR	Sondeur Atmosphérique du Profile d'Humidité Intertropicale par Radiométrie (Megha-Tropiques)
SAR	Synthetic Aperture Radar
SARRA-H	Système d'Analyse Régional des Risques Agroclimatiques-Habillé
SBA	Split-Based Approach
SCA	Scatterometer (ESA)
SCaMPR	Self-Calibrating Multivariate Precipitation Retrieval (NESDIS)
ScaRaB	Scanner for Radiation Budget (Megha-Tropiques)
scPDSI	Self-Calibrated PDSI
SCS	Single-Look Complex Slant Products
SCSMEX	South China Sea Monsoon Experiment
SCSMEX/NESA	SCSMEX Northern Enhanced Sounding Array
SCSMEX/SESA	SCSMEX Southern Enhanced Sounding Array
SD (1)	Snowfall Detection
SD (2)	Standard Deviation
SDCI	Scaled Drought Condition Index
SDG	Sustainable Development Goal
SDII	Simple Daily Intensity Index (ETCCDI)
SEA	Southeast and East Asia
SEAK	Southeast Alaska
SEM	Semiempirical Model

SEVIRI	Spinning Enhanced Visible and Infrared Imager (EUMETSAT)
SFI	Seeder-Feeder Interactions
SFR	Snowfall Rate (NOAA)
SH	Southern Hemisphere
SI	Snow Index
SIA	Sea Ice Age
SIC	Sea Ice Concentration
SID	Sea Ice Drift
SIDOC	SAR Images Dark Object Classifier
SIT	Sea Ice Thickness
SLH	Spectral Latent Heating
SM	Soil Moisture
SMA	Soil Moisture Anomaly
SMAP	Soil Moisture Active Passive (NASA)
SMMR	Scanning Multichannel Microwave Radiometer (NOAA)
SMOS	Soil Moisture and Ocean Salinity (ESA)
SM2RAIN	Soil Moisture to Rain Algorithm (CNR-IRPI)
S-NPP	Suomi National Polar-Orbiting Partnership (NASA/NOAA)
SNR	Signal-to-Noise Ratio
SON	September-October-November
SOS	Start of Season
SPA	Specific Power Attenuation
SPCZ	South Pacific Convergence Zone
SPEI	Standardized Precipitation Evapotranspiration Index
SPI	Standardized Precipitation Index
SPICE	Solid Precipitation Intercomparison Experiment (WMO)
SPoRT	Short-term Prediction Research and Transition Center (NASA)
SPP (1)	Satellite Precipitation Product
SPP (2)	Seasonal Performance Probability
SPS	Special Weather Statement (NWS)
SPURS	Salinity Processes in the Upper Ocean Regional Study
SR	Success Ratio
SRE	Satellite-Based Rainfall Estimate
SREM2D	Two-Dimensional Satellite Rainfall Error Model
SRI	Surface Rainfall Intensity
SRT	Surface Reference Technique
SSA (1)	Space Situational Awareness
SSA (2)	Sub-Saharan Africa
SSN	Spatial Stream Network
SSM/I	Special Sensor Microwave/Imager (DMSP)
SSMIS	Special Sensor Microwave Imager Sounder (DMSP)
SSM/T	Special Sensor Microwave Temperature (DMSP)
SSP	Surface Salinity Profiler

SSRGA	Self-Similar Rayleigh-Gans Approximation
SSS	Sea Surface Salinity
SST	Sea Surface Temperature
SSU	Stratospheric Sounding Unit (NOAA)
SSW	Sea Surface Wind
STAR	Center for Satellite Applications and Research (NOAA-NESDIS)
STDV	Standard Deviation
STH	Storm-Top Height
STIa	South Tropical Indian Area
STPa	South Tropical Pacific Area
SWA	South and West Asia
SWB	Soil Water Balance
SWE	Snow Water Equivalent
SWER	Snow Water Equivalent Rate
SYNOP	Surface Synoptic Observation (WMO)
TAMSAT	Tropical Applications of Meteorology Using Satellite and Ground-Based Observations
TAPEER	Tropical Amount of Precipitation with an Estimate of Errors
TARCAT	TAMSAT African Rainfall Climatology and Time Series
TB	Brightness Temperature
TC	Tropical Cyclone
TCA	Triple Collocation Analysis
TCC	TRMM Composite Climatology
TC4	Tropical Composition, Cloud and Climate Coupling Experiment
TDTS	Time-Dependent Two-Stream Method
TEMPEST	Temporal Experiment for Storms and Tropical Systems
TEMPEST-D	TEMPEST Technology Demonstration
TIR	Thermal Infrared
TLC	Tropical-Like Cyclone
TMD	Thai Meteorological Department (Thailand)
TMI	TRMM Microwave Imager
TPMA	TRMM Multisatellite Precipitation Analysis (NASA)
TOOCAN	Tracking Of Organized Convection Algorithm through a 3-D segmentation
TOGA-COARE	Tropical Ocean Global Atmosphere Coupled Ocean-Atmosphere Response Experiment
TOVS	TIROS Operational Vertical Sounder
TPW	Total Precipitable Water
TQV	Total Precipitable Water Vapor
TRMM	Tropical Rainfall Measuring Mission
TROPICS	Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (NASA)

TSX	TerraSAR-X (DLR)
TVA	Tennessee Valley Authority
TWP-ICE	Tropical Warm Pool – International Cloud Experiment
TWSA	Terrestrial Water Storage Anomaly
T2M	Two-Meter Temperature
UCA	Urban Climate Archipelago
UCLM	Universidad de Castilla-La Mancha
UH	University of Helsinki
UHI	Urban Heat Island
UMORA	Unified Microwave Ocean Retrieval Algorithm
UN	United Nations
UNEP	United Nations Environment Programme (UN)
UNESCO	United Nations Educational, Scientific and Cultural Organization (UN)
US	United States of America
USCRN	US Climate Reference Network
USDA	US Department of Agriculture
USDM	US Drought Monitor
USGS	US Geological Survey
UTC	Universal Time Coordinated
UWSI	Urban Water Stress Index per Individual
VAM	Vulnerability Assessment and Mapping (WFP)
VarBC	Variable Bias Correction
VI	Vegetation Index
VMI	Vertical Maximum Intensity
VN	Validation Network (GPM)
WATCH	Water and Global Change (EU)
WCOM	Water Cycle Observation Mission (CMA)
WCRP	World Climate Research Programme (WMO)
WDCC	World Data Center for Climate (DKRZ)
WDM6	WRF Double-Moment 6-Class
WEA	Western East Africa
WFF	Wallops Flight Facility (NASA)
WFO	Weather Forecast Office (NWS)
WFP	World Food Programme
WFDEI-CRU	WATCH Forcing Data ERA Interim-CRU
WII	Weather Index-Based Insurance
WMMD	Wet Millimeter Days
WMO	World Meteorological Organization (UN)
WR	Weather Radar
WRF	Weather Research and Forecasting Model
WRF-ARW	WRF-Advanced Research WRF
WRSI	Water Requirements Satisfaction Index
WS	Wind Speed

WSR-57	Weather Surveillance Radar 1957 (NWS)
WV	Water Vapor
XCAL	Intersatellite Calibration Working Group (GPM)
ZAR	Zones À Risque model
2DVD	2D Video Disdrometer